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Design of a Memory System for Sparse Data Processing

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THESIS

DESIGN OF A MEMORY SUB-SYSTEM FOR SPARSE DATA PROCESSING

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Acknowledgements

TODO

"Work. Or school. This is for the money. But what is anything else, for that matter, but required means of enhancing your bare living – better sleep and better food – like spending decades just to turn a little knob higher on a pathetic radio called existence."

Karl Kristian Flores, The Goodbye Song.

Résumé

TODO

Abstract

TODO

List of Publications

This thesis is based on the following publications:

I Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. Dedicated Hardware Accelerators for Processing of Sparse Matrices and Vectors: A Survey. *ACM Trans. Archit. Code Optim.* 21, 2, Article 27 (Feb. 2024), 26 pages. Reference [28]

II Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. SpDCache: Region-Based Reduction Cache for Outer-Product Sparse Matrix Kernels. In *2024 IEEE 35th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. IEEE Computer Society, Los Alamitos, CA, USA, 3–7. Reference [29]

III **TODO**

The papers will be referred to in the thesis using their Roman numerals.

Summary

Notations

$$\forall(a,b)\in\mathbb{Z}^2, [\![a,b]\!]=[a,b]\cap\mathbb{Z}$$

nnz, M, A, b and c, $\overline{nnz_r}$, d_r ...

TODO

Glossary

band Refers to an area of a matrix that is concentrated around the diagonal. 9, 10, 17, 18, 19, 39, 42, 43, 44, 47, 48, 49, 54, 55, 56, 57, 58, 59, 60, 61, 62, 86

banded Refers to a matrix structure where the non-zero elements are concentrated around the diagonal. 17, 18, 19, 29, 32, 34, 35, 39, 40, 42, 47, 48, 49, 51, 54, 56, 57, 59, 61, 62, 86

bandwidth With this typography, refers to the size of the band of a matrix. 17

common rank Refers to a rank that is common between two data-entries, when computing a kernel. 20, 21, 26, 28

dense format Is the uncompressed format where all zeros and non-zeros are stored, without any use of metadata. 18, 25, 27

eviction Is a cache event where a valid cache entry is flushed and freed. 36, 37, 38, 39, 40, 41, 42, 45, 47, 50, 51, 52, 53, 54, 55, 56, 57, 59, 62, 64, 86

fiber Is the generic term to describe a one-dimensional stream of data or metadata. It corresponds to a row or column in the case of a two-dimensional matrix. A stream of indices is a *fiber* of metadata. 24, 28

gather Refers to reading data from memory. We specifically use this term to categorize data transfers that are irregular. 28, 32

hit Is a cache event where a given address does match one of the valid cache entries. 36, 37, 38, 41, 42, 53, 55, 59, 64

ineffectual operation Is an operation that leads to a neutral result, 0 for an addition or 1 for a multiplication. 24

intersection Refers to the process of finding non-zero partial sums, when computing a matrix multiplication. 21, 22, 23, 24, 25, 26, 28, 31, 32, 89

irregular access See "random" access. 9, 15, 16, 32, 34, 35, 36, 49, 50

irregular data Refers to data that has low spatial and temporal locality, thus being more likely to generate irregular accesses. 32, 36, 37

metadata Is data that provides information about other data, such as index arrays of sparse formats. 9, 10, 18, 20, 24, 25, 28, 79

miss Is a cache event where a given address does not match any of the valid cache entries. 26, 29, 36, 37, 38, 41, 42, 53, 64

partial sum Refers to the intermediary result before updating the output, when computing a matrix multiplication. 9, 21, 25, 26, 28

'perfectly' banded Refers to a matrix structure where all the non-zero elements are concentrated within a delimitated area around the diagonal (the band). 17, 49, 52, 54, 55, 56, 57, 60

"random" access Is a memory access that occurs on an arbitrary memory address (not necessarily conscutive in memory with previous and next accesses). 9, 10, 16

"random" update Is a "random" access where the access is a read, modify and write on a given address. It is similar to a "random" *reduction*. 26

rank Refers to a dimension of the kernel. 9, 18, 20, 21, 22, 23, 24, 28

reduction Refers to the process of updating an entry. 10, 21, 26, 27, 28, 29, 31, 32, 34, 35, 36, 37, 38, 40, 41, 42, 43, 44, 46, 47, 58, 86, 89

reduction cache Is a data-cache supporting *reduction* operations. 37, 38, 39, 40, 42, 43, 44, 45, 47, 48, 49, 50, 51, 53, 54, 55, 56, 57, 61, 62, 64

scatter Refers to updating data in memory. We specifically use this term to categorize data transfers that are irregular. 28, 32

sequential accesses Are a group of memory accesses (from a data array, for example) that occur on conscutive memory addresses, in order through time. 18, 36

set Refers to a fixed number of cache-lines, a subdivision related to set-associative caches. 10, 36, 38, 40, 43, 45, 46, 49, 52, 53, 59, 67

sparse format Is a compression format for sparse matrices, which avoids storing zeros by using metadata. 10, 18, 19, 20, 22, 23, 24, 25, 26, 32, 38, 79

tag Refers to cache metadata that allows identification of cache-line memory addresses. 36

tile Refers to a sub-region of a *tiling* process. 22, 23, 24, 31

tiling Is a method consisting in splitting data into several independent sub-regions to be processed separately. 10, 22, 23, 24, 26, 28, 31, 32, 79, 86, 89

way Refers to the number of cache-lines that a *set* contains. 36, 38, 43, 49, 59

Acronyms

BaCSC *Banded Compressed Sparse Column*. 19, 32, 86

BCSR *Blocked Compressed Sparse Row*. 19

COO *COOrdinate*. 18, 19, 20, 27, 32, 79, 86

CSC *Compressed Sparse Column*. 18, 19, 20, 24, 25, 26, 27, 28, 32, 38, 50, 79, 86, 89

CSR *Compressed Sparse Row*. 18, 19, 20, 23, 24, 25, 26, 27, 28, 32, 79, 86, 87, 89

DIA *DIAgonal*. 19

DRC *Dense Region Cache*. 40, 41, 42, 43, 44, 45, 46, 47, 54, 55, 59, 60, 61, 62, 67, 86

ELL *ELLPACK*. 19

GC Generic Cache. 42, 43, 44, 45, 64, 67

GRC *Generic Region Cache*. 40, 42, 43, 44, 45, 46, 47, 50, 62, 67, 86

`Gust` *Gustavson*. 20, 21, 22, 23, 26, 28, 29, 31, 32, 88, 89

HYB *HYBbrid*. 19

IBR *In-Band Region*. 40, 41, 42, 47

`IP` *Inner-Product*. 20, 21, 22, 23, 24, 25, 26, 28, 29, 31, 32, 52, 88, 89

LRU *Least Recently Used*. 36, 43, 50, 60

MM Matrix-Matrix. 20, 21, 22, 23, 28, 32, 89

MV Matrix-Vector. 20, 21, 22, 23, 24, 28, 32, 89

OBR *Out-of-Band Region*. 40, 41, 42, 47

`OP` *Outer-Product*. 20, 21, 22, 23, 24, 26, 27, 28, 29, 31, 32, 34, 35, 38, 39, 40, 42, 44, 48, 50, 51, 52, 54, 56, 61, 62, 86, 88, 89

PO *partial output*. 21, 22, 23, 26, 28, 32, 45

RedC Reduction Cache. 42, 43, 44, 45, 46, 64, 67

RMW *Read-Modify-Write*. 41, 42, 43, 44, 58, 59

RMWQ *Read-Modify-Write Queue*. 41, 42

SpDC SpDCache. 35, 39, 40, 41, 42, 43, 44, 45, 46, 47, 49, 50, 54, 55, 58, 60, 61, 62, 64, 67, 86

SpGEMM General Sparse Matrix-Matrix multiplication. 20

SpGEMV General Sparse Matrix-Vector multiplication. 20

SpMM Sparse Matrix-Matrix multiplication. 20

SpMSpM Sparse Matrix-Sparse Matrix multiplication. 20, 22, 24, 25, 26, 27, 28, 29, 31, 32, 88, 89

SpMSpV Sparse Matrix-Sparse Vector multiplication. 20

SpMV Sparse Matrix-Vector multiplication. 20, 23, 24, 31, 32, 34, 35, 38, 39, 40, 42, 44, 48, 50, 51, 52, 54, 56, 61, 62, 86, 89

SRT *Sparse Region Table*. 40, 41, 42, 43, 44, 45, 46, 47, 61, 62, 67, 86

Introduction

The rapid growth of data-intensive applications has made sparse and irregular data structures increasingly prevalent in scientific computing, machine learning, and graph analytics. However, efficiently processing sparse matrices remains a fundamental challenge due to their inherent irregularity, which leads to unpredictable memory access patterns, limited opportunities for data reuse, and suboptimal utilization of hardware resources. Existing hardware accelerators have demonstrated partial solutions, yet they often rely on restrictive assumptions, exhibit limited scalability, or fail to fully exploit the memory bandwidth.

This thesis addresses the following central question:

PROBLEM STATEMENT

How can hardware and architectural mechanisms be designed to efficiently support computation with irregular sparse data, in particular random reductions, while ensuring scalability, adaptability, and high performance across diverse workloads?

To tackle this problem, we investigate the limitations of existing algorithms and accelerators, propose a novel cache architecture with support for reduction operations, tailored to sparse data. We then evaluate the behaviour of this specialized cache through modeling, simulation, and hardware prototyping.

This manuscript is structured around several key contributions. We begin with an in-depth discussion of irregular data, with a particular emphasis on sparse matrices (Chapter 1). This includes a study of fundamental sparse matrix multiplication kernels and their associated algorithms, highlighting the main hardware challenges that arise when dealing with sparse data. A high-level analysis of algorithms for sparse data computing is provided, with a focus on basic storage schemes and opportunities for data reuse. Building on this foundation, we review the state-of-the-art in hardware accelerators designed for efficient sparse matrix computing (Chapter 2). This review identifies common characteristics, proposes a classification of existing accelerators and design tendencies, offers a comparative analysis, and identifies the major tendencies across designs. This analysis of the state of the art guided our subsequent architectural choices.

Following this survey, we introduce a novel cache architecture specifically designed to address the challenge of "random" reductions (Chapter 3). The proposed cache integrates reduction support and organizes its memory into multiple regions optimized for denser and sparser data regions. A Python model was developed to simulate memory transactions in this cache and to compare its behavior with that of a generic cache. To further investigate its performance, we complement this analysis with a probabilistic model, implemented in C, that enables rapid simulation and sizing parameters exploration depending on matrices structure (Chapter 4). At the micro-architectural level, we detail the virtual partitioning of cache memory into multiple configurable regions, which ensures both flexibility and adaptability to diverse workloads (Chapter **??**).

The proposed cache is then evaluated through RTL simulation, which reveals two distinct classes of matrices that exhibit different performance profiles (Chapter 6). Finally, the manuscript outlines potential future enhancements to improve both performance and scalability, including a multicore extension of the cache architecture and optimizations targeting bandwidth efficiency (Chapter 7).

Contributions of the Thesis

The main contributions of this thesis can be summarized as follows:

1. **Analysis of irregular and sparse data computing** (Chapter 1)
   - Study of fundamental sparse matrix multiplication kernels and algorithms.
   - Identification of the key hardware challenges associated with sparse data processing.
   - High-level analysis of algorithms for sparse data computing, with emphasis on storage schemes and data reuse opportunities.
2. **Survey and classification of state-of-the-art accelerators** (Chapter 2)
   - Comprehensive review of hardware accelerators for sparse matrix computing.
   - Identification of common architectural features and design tendencies.
   - Comparative analysis to explain design choices under varying objectives and constraints.
3. **Design of a cache architecture for random reductions** (Chapter 3)
   - Proposal of a cache with native reduction support, capable of handling both dense and sparse data via multiple cache regions.
   - Development of a Python-based simulation model to analyze memory transactions and benchmark against a generic cache.
4. **Probabilistic modeling and exploration of cache behavior** (Chapter 4)
   - Implementation of a lightweight C-based probabilistic model for rapid simulation.
   - Exploration of cache dimensioning and behavior across diverse workloads.
5. **Micro-architecture of the reduction cache** (Chapter **??**)
   - Introduction of a virtual memory partitioning scheme enabling multiple configurable regions.
   - Emphasis on flexibility and adaptability in cache design.
6. **Evaluation through RTL simulation** (Chapter 6)
   - Assessment of the proposed cache architecture using RTL-level simulation.
   - Identification of two distinct groups of sparse matrices leading to different performance profiles.
7. **Future directions for performance and scalability** (Chapter 7)
   - Proposal of a multicore version of the cache to extend scalability.
   - Exploration of bandwidth optimization strategies.

Chapter 1

Background

Contents

1.1 Introduction

MODERN computing systems are designed to deliver high performance by exploiting memory hierarchies, parallelism, and increasingly sophisticated hardware features. However, the efficiency of these systems is strongly tied to the regularity of memory access patterns. Applications with predictable, sequential data access can fully benefit from cache locality, prefetching mechanisms, and vectorization. In contrast, applications characterized by irregular accesses, where memory locations are not known in advance or follow non-contiguous patterns, are not able to achieve peak performance.

In domains such as graph analytics, sparse linear algebra, unstructured mesh computations, and database queries, the memory access patterns are often irregular. In these applications, pointer chasing, indirect indexing, and dynamically changing data structures disrupt spatial and temporal locality. As a result, the hardware struggles to hide the latency of memory accesses, leading to poor cache utilization and underutilization of compute ressources.

The impact of irregular accesses extends beyond raw performance. They also complicate algorithm design, hinder scalability on many-core architectures, and increase energy consumption due to repeated off-chip memory requests. Understanding and addressing these challenges is therefore a central concern in both computer design and application optimization. This chapter provides the necessary background for analyzing irregular accesses, highlighting their sources, effects, and the architectural considerations they expose, focusing on the case of indirect indexing in sparse matrix computation. The content of this chapter is partially based on the content from Papers I and II.

DEFINITION

Irregular accesses, or **"random" accesses**, are sequences of accesses where the sequence can not be easily predicted.

1.2 Sparse Matrices

SPARSE matrices are two dimensional arrays filled with many zeros and stored in compressed formats in memory. These sparse matrices are commonly used in diverse fields such as linear algebra, where sparse matrix multiplication is used to solve linear systems of equations for scientific purposes or simulation, as well as graph analytics where the vertices of a graph are represented as a sparse matrix [10], or finally transformers or other similar AI features where weights and activations can be represented and processed as sparse matrices [17, 20]. In these contexts, matrices are large, with dimensions up to billions of elements, and highly sparse, with non-zero densities down to 10^{-7} [35].

We note matrices with capital letters A, B, C, conversely to vectors a, b, c. Their dimension are denoted using M, N and K. For example, we can define a square matrix A of dimension M, or a matrix B of dimension (K, N). We note nnz the number of non-zero elements of a matrix, and d the associated density. With the example of a matrix $A(M, N)$:

$$d_A = \frac{nnz_A}{M \times N}$$

DEFINITION

A **sparse matrix** is a matrix filled with many zeros.

NOTE

In general, we consider a matrix to be highly sparse when its number of non-zero elements has the same order of magnitude than its dimension, $nnz \sim M$. But there is no intrinsic definition of the frontier between a sparse and a dense matrix. In this manuscript, we consider a matrix as sparse when it needs to be compressed in memory (see Section 1.3).

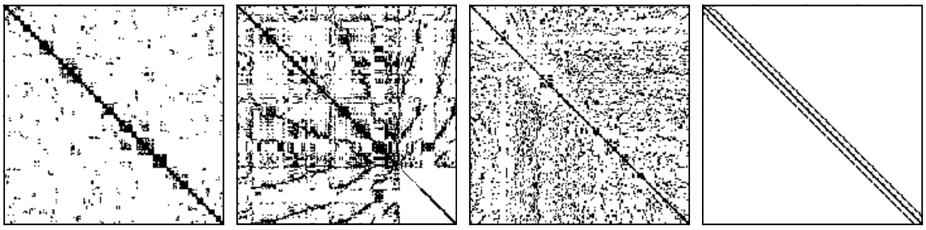
1.2.1 Banded Matrices

Matrices used in scientific computation originate from physical modelling where a system of partial differential equations is discretized, resulting in a large sparse matrix. In these systems, the forces between elements decreases quickly with their distance, creating banded matrices. Moreover, numerical techniques exist to transform matrices into banded matrices [38, 39, 55].

A banded matrix is a matrix which has a dense region around its main diagonal and which is sparser away from the diagonal. To give a few visual examples, Figure 1.1 shows the structure plots for four typical matrices found in the SuiteSparse Matrix Collection [35], having a high-density band along the diagonal. In Figure 1.2, we illustrate a banded, square matrix of dimension M. We define w_B as the width of the banded region, or *bandwidth*, as shown in *red* in the figure. Sometimes it is easier to refer to the half-*bandwidth*, w_{hB}, which respects the relation $w_B = 2 \times w_{hB} - 1$. The average density in the band and out of the band is defined as d_{IB} and d_{OB}, respectively. As in Figure 1.1d, in a 'perfectly' banded matrix, there are no elements outside the band, thus $d_{OB} = 0$.

DEFINITION

A **banded matrix** is a sparse matrix which has a higher density of non-zero elements around the diagonal. A **'perfectly' banded matrix** concentrates all its non-zero elements within a



(a) *roadNet-CA* **(b)** *scircuit* **(c)** *msc10848* **(d)** *raefsky3*

Figure 1.1: Examples of Banded Matrix Structure

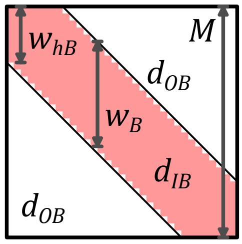


Figure 1.2: Banded Matrix Definition

delimitated region around the diagonal. This region is called the band.

NOTE

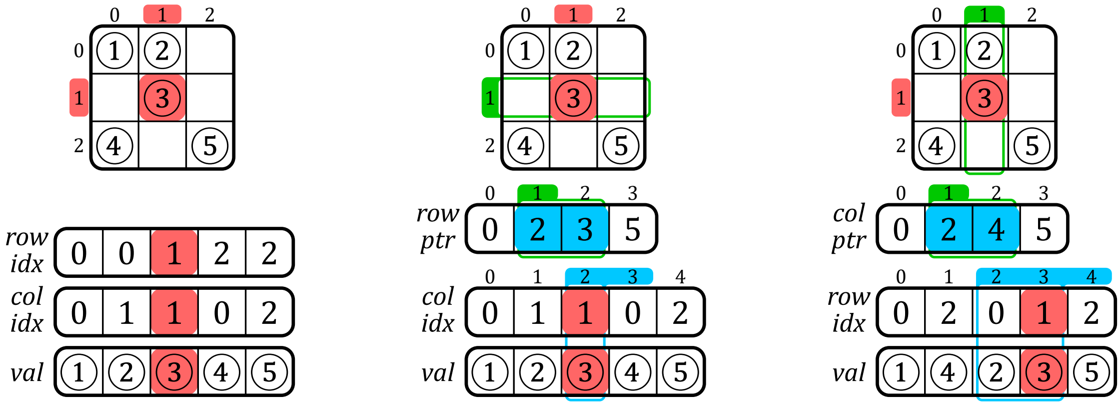
The SuiteSparse Matrix Collection [35] is a large, open source data base of sparse matrices from multiple real-world applications, many of which were provided by companies such as Boeing. It gathers around 3,000 matrices of various sizes, densities and structure, which are extensively used for benchmarking (see Chapter 2).

1.3 Sparse Formats

IN the absence of a compressed format, matrices are stored in fixed-size two-dimensional arrays. With this dense format, elements can be randomly accessed ($\mathcal{O}(1)$), and sequential accesses are highly efficient. The two dimensions are called ranks which correspond to the rows and columns. For matrices, there are thus two variants of the dense format, depending on which rank is stored sequentially in memory. In a *row-wise* (or *row-major*) storage, the elements within the rows are sequential, and conversely for *column-wise* (or *column-major*).

Sparse matrices contain a very large number of zeros, and to avoid storing these repeated zeros, there exist many compressed formats [6, 7, 13, 32, 46, 52], where certain formats are best-suited for matrices with a specific structure. The main principle of these sparse formats is to avoid storing the zero elements, but this requires additional metadata which gives the position of the non-zero elements.

For example, let us consider the commonly used *COOrdinate* (COO) format that is relatively simple. It consists of three tables of size nnz with only the non-zero elements (val table in Figure 1.3a) and their row and column indices (row and col idx tables). The tuples do not necessarily need to be sorted, which makes it easy to append new entries. However, if the tuples are not sorted, this format is not well suited for sequential traversal or locating a specific matrix entry. This format can be sorted *row-wise* as in the example in Figure 1.3a or *column-wise* if needed, but in this case, it is less efficient than the two other widely used sparse formats *Compressed Sparse Row* (CSR) (Figure 1.3b) and its *column-major* counterpart *Compressed Sparse Column* (CSC) (Figure 1.3c), which are sorted by construction. With CSR, the val table of size nnz stores the non-zero elements sorted *row-wise* and the idx table of size nnz stores the corresponding column indices, similarly to COO. The third table ptr of size $M+1$ stores the positions of the rows. For example, in Figure 1.3b, the second row of index 1 (*green*) has one non-zero element between positions 2 and 3 (*blue*, 3 excluded) of tables idx and val. The CSC format works similarly but in a *column-wise* manner: the val table of size nnz is sorted *column-wise* and the idx table of size nnz stores column indices. The ptr table of size $N+1$ stores the positions of the columns. In the example in Figure 1.3c, the second column of index 1 (*green*) has non-zero elements between positions 2 and 4 (*blue*, 4 excluded) of tables idx and val. The ptr table allows CSR and CSC to have a lower memory footprint than COO.



(a) COO **(b)** CSR **(c)** CSC

Figure 1.3: Example of a Sparse Matrix in COO, CSR and CSC Formats

Many others classes of sparse formats exists such as *Blocked Compressed Sparse Row* (BCSR), *DIAgonal* (DIA), *ELLPACK* (ELL) and *HYBbrid* (HYB), which all have their own advantages depending on the matrix structure or the hardware setup. Additional information about these other formats is provided in Appendix A.

In general, there are many derivates from the above main categories of sparse formats. Ultimately, we can create any format we want depending on the software and hardware constraints. Later in this manuscript, we propose our own custom sparse format called *Banded Compressed Sparse Column* (BaCSC) that is an extension of CSC for banded matrices. As shown in Figure 1.4, we added extra information in the ptr table to identify the beginning and end of the band. Indeed, three pointers instead of one are now needed per column, as shown with the example of column 2 in *yellow*. The column is split in one in-band (IB) region in the middle and two out-of-band (OB) regions around. The ptr table size thus becomes $3 \times M + 1$. With this format, we can avoid the computational cost of identifying in which region non-zero elements are, while traversing the matrix, a feature we take advantage of in Chapter 6.

(MOVE BaCSC in future chapter evaluation)

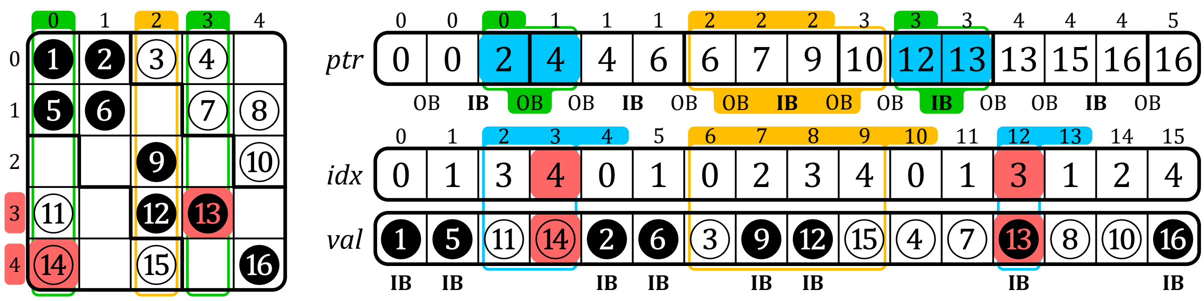


Figure 1.4: Example of a Sparse Matrix in BaCSC Format

DEFINITION

A **sparse format** is a memory representation of a sparse matrix which avoids storing zeros by using metadata.

NOTE

The most commonly used sparse formats are COO, CSR and CSC.

CONCLUSION

With the use of sparse formats, it is possible to greatly **decrease the memory footprint** of a sparse matrix while increasing the data dimensions. These formats **can be tailored** based on the needs of the software and hardware.

1.4 Algorithms for Dense and Sparse Matrix Multiplication

THE main matrix multiplication kernels that are of interest are Matrix-Vector (MV) and Matrix-Matrix (MM), which include SpMV and SpMSpV ($A \times b = c$), SpGEMV ($A \times b + d = c$), SpMM and SpMSpM ($A \times B = C$), and SpGEMM ($A \times B + D = C$), with 'Sp' for *sparse* and 'GE' for *general*. SpMV, as well as its variant SpGEMV, are by far the dominant operations in modern algebraic kernels (linear solvers, eigensolvers) which have been explicitly based on them [15, 22, 25, 30, 34, 43, 60]. Since these kernels are ubiquitous, their efficient implementation is primordial. In contrast, the SpMSpM multiplication is less frequent in standard algebraic kernels, but it is extensively used in algebraic graph manipulation [3, 9, 12, 14, 21, 31, 44], as well as in multigrid solvers [2, 8, 37, 45]. The growing importance of large graphs has motivated new research on optimizing SpMSpM applications.

In the following sections, we focus on SpMV and SpMSpM, with the notation $A(M, K) \times b(K) = c(M)$ and $A(M, K) \times B(K, N) = C(M, N)$, respectively. The rank K is the common rank between the two inputs A and B (or b). The existence of different ranks allows several algorithmic possibilities to compute matrix multiplication: the *Inner-Product* (`IP`) algorithm, the *Outer-Product* (`OP`) algorithm and *Gustavson*'s (`Gust`) algorithm [16, 24]. In all cases, the matrix A is traversed sequentially. The three algorithms, however, impose different access patterns for B (or b) and C (or c). Indeed, in many cases, they have to be accessed multiple times, as discussed in the following sections.

NOTE

In this manuscript, we focus on only two kernels, SpMV and SpMSpM. Note that, among MV kernels, SpMV is the most used and we specifically work on this kernel in Chapter 6. SpMSpM is also widely used and studying this MM kernel is of interest as the memory access patterns in both matrices are irregular.
The *general* ('GE') version of these kernels is not fundamentally different, as the majority of the compute operations are in the matrix multiplication, so going forward, we will ignore the 'GE' variants.

1.4.0.1 Inner-Product Algorithm

The *Inner-Product* algorithm [16] is an intuitive way to calculate a matrix multiplication because it follows the loop order of the mathematical formulas (1.1) and (1.2) for MV and MM, respectively. The common rank K is the innermost loop of the algorithm (see Algorithms 1.1 and 1.2, lines 2 and 3, respectively). The output C (or c) is always updated sequentially, but the B-matrix (or b-vector) will be read entirely M times. In the context of sparse matrices, the reads of B (or b) are conditioned by the non-zero elements of A, which requires indirect accesses. If B (or b) is also sparse, the algorithm needs to perform *intersections* between the non-zero elements in the rows of A and the columns of B (or in the b-vector) (see Section 1.5). In Algorithm 1.2, exchanging M and N (lines 1 and 2) simply causes the output to be updated *column-wise*.

$$\forall i \in [\![0, M-1]\!],\ c_i = \sum_{k=0}^{K-1} A_{i,k} \times b_k \tag{1.1}$$

$$\forall (i,j) \in [\![0, M-1]\!] \times [\![0, N-1]\!],\ C_{i,j} = \sum_{k=0}^{K-1} A_{i,k} \times B_{k,j} \tag{1.2}$$

Algorithm 1.1: Dense Matrix-Vector *Inner-Product* Algorithm

1 **for** $i \in [\![0, M-1]\!]$ **do** // row first
2   **for** $k \in [\![0, K-1]\!]$ **do**
3         $c_i \mathrel{+}= A_{i,k} \times b_k$

Algorithm 1.2: Dense Matrix-Matrix *Inner-Product* Algorithm

1 **for** $i \in [\![0, M-1]\!]$ **do** // A-row first
2   **for** $j \in [\![0, N-1]\!]$ **do**
3   **for** $k \in [\![0, K-1]\!]$ **do**
4             $C_{i,j} \mathrel{+}= A_{i,k} \times B_{k,j}$

1.4.0.2 Outer-Product Algorithm

The *Outer-Product* algorithm [16] has the opposite rank loop order than the *Inner-Product*. Indeed, Algorithms 1.3 and 1.4 show that the common rank K is the outermost loop (line 1). It changes nothing mathematically except that the B-matrix (or b-vector) is always read sequentially. In the context of sparse matrices, the output C (or c) is updated with indirect accesses. The algorithm needs to perform *reductions* of the partial sums $A_{j,k} \times B_{k,j}$ (or $A_{j,k} \times b_k$), as we show in Section 1.5. To avoid irregularity when updating C (or c), the **OP** algorithm is often separated into two phases. First, during the *multiply* phase, we compute several *partial outputs* (*POs*) consisting of intermediate matrices (or vectors) of partial sums $A_{j,k} \times B_{k,j}$ (or $A_{j,k} \times b_k$), then during the *merge* phase, we accumulate all the *POs* to form the final C-matrix (or c-vector). With this method, **OP** necessitates the generation of at most K *POs* of densities conditioned by the non-zeros elements of both inputs A and B (or b).

Algorithm 1.3: Dense Matrix-Vector *Outer-Product* Algorithm

1 **for** $k \in [\![0, K-1]\!]$ **do** // column first
2   **for** $i \in [\![0, M-1]\!]$ **do**
3         $c_i \mathrel{+}= A_{i,k} \times b_k$

Algorithm 1.4: Dense Matrix-Matrix *Outer-Product* Algorithm

1 **for** $k \in [\![0, K-1]\!]$ **do** // common rank first
2   **for** $i \in [\![0, M-1]\!]$ **do**
3   **for** $j \in [\![0, N-1]\!]$ **do**
4             $C_{i,j} \mathrel{+}= A_{i,k} \times B_{k,j}$

1.4.0.3 Gustavson's Algorithm

As the MM algorithms have three ranks instead of two for MV, there is a third loop ordering, known as *Gustavson*'s algorithm [24] where the common rank K is the middle-loop (see line

2 in Algorithm 1.5). With this approach, the output is computed row-by-row with *PO*-rows generated and accumulated for each output C-row. With this algorithm, updates of the C-rows are sequential, but elements within the rows are updated with indirect accesses. In the dense case, the B-matrix is entirely read M times and K *PO*-rows, of size N, are generated. In the context of sparse matrices, conversely to the C-rows, the selection of the rows in B is indirect, but the accesses are sequential within a row. Thus, the *intersection* search is only needed for the B-rows and not each element within rows. The *PO*-rows can be accumulated before computing the next C-row. Overall, this algorithm can be seen as a trade-off between **IP** and **OP** for MM kernels, since the indirect accesses are shared between B and C. Again, for this algorithm, exchanging M and N causes the three matrices to be accessed *column-wise* instead of *row-wise*.

Algorithm 1.5: Dense Matrix-Matrix *Gustavson*'s Algorithm

```
for i ∈ ⟦0, M − 1⟧ do
    for k ∈ ⟦0, K − 1⟧ do        // common rank
        for j ∈ ⟦0, N − 1⟧ do
            C_{i,j} += A_{i,k} × B_{k,j}
```

1.4.0.4 Comparison of the Algorithms

In Table 1.1, we summarize the main features of **IP**, **OP** and **Gust** in terms of accesses to the matrix A, B (or b) and C (or c). From this high level analysis, we can easily see that the data that is accessed with indirections shifts depending on the algorithm, from the input B (or b) to the output C (or c), with **Gust** being a trade-off between **IP** and **OP** for MM multiplication. To have a better understanding of the consequences on the actual number of memory accesses, we propose a more detailed analysis in Section 1.5.4, for the case of SpMSpM.

NOTE

Note that these algorithms are available in dense as well as in sparse MV and MM multiplications. In the sparse variants, the above algorithms (from 1.1 to 1.5) have the same behavior from a mathematical point-of-view, but need adjustments to traverse a sparse format instead of a dense matrix. Sparse versions of these algorithms are presented in Section 1.5.

CONCLUSION

There are *three main algorithms* to compute sparse MV and MM kernels, ***Inner-Product*** (**IP**), ***Outer-Product*** (**OP**) and ***Gustavson*** (**Gust**). Each presents different characteristics, the main one being the **irregularity of the accesses**: on the **input** matrix B (or vector b) with **IP**; on the **output** matrix C (or vector c) with **OP**; on both with **Gust** but at **row scale**.

1.4.1 Tiling

The *tiling* process [23] consists in partitioning a matrix: a matrix may be divided into *tiles*, non-overlapping sub-matrices. In other words, this method consists of adding ranks to delimit the *tiles*, and thus adding loop iterations in the kernel algorithm.

Tiling methods make it possible to process smaller portions of the matrix and thus to adapt the working set to the size of the local memory [36, 56], particularly when the matrix has a

Table 1.1: Comparison of Matrix Multiplication Algorithms

	Inner-Product (`IP`)[1]	***Outer-Product*** (`OP`)[1]	***Gustavson*** (`Gust`)[2]
A(M,K) **Accesses**	• *row-wise* • read N times[3,4] • sequential	• *column-wise* • read once • sequential	• *row-wise* • read once • sequential
B(K,N) **Accesses**	• *column-wise* • read M times[5] • not sequential	• *row-wise*[6] • read $M \times d(A)$ times[4,7] • sequential	• *row-wise* • read $M \times d(A)$ times[7] • sequential within rows
C(M,N) **Accesses**	• generated *row-wise*[6] • one sequential traversal	• generated without specific structure • K *POs* produced[8] (each *PO* has $\overline{nnz_c}(A) \times \overline{nnz_r}(B)$ non-zero elements[9])	• generated *row-wise* • $K \times d(A)$ *PO*-rows produced[10] for each row of A (M) ($\overline{nnz_r}(B)$ non-zero elements per *PO*-row)

[1] Considering $N = 1$ in the case of MV multiplication.
[2] Only for MM multiplication.
[3] Or less than N times if B has empty columns.
[4] Read once in the case of MV multiplication.
[5] If B is stored with a sorted sparse format, else it would be read $\overline{nnz_r}(A) \times M = nnz(A)$ times.
[6] *Element-wise* in the case of MV multiplication (there is no row).
[7] Which is equal to $\overline{nnz_c}(A)$, the average number of non-zero elements in the columns of A.
[8] Or less than K times if A has empty columns or B has empty rows.
[9] $\overline{nnz_c}(A)$ in the case of MV multiplication.
[10] Which is equal to $\overline{nnz_r}(A)$, the average number of non-zero elements in the rows of A.

structure with denser regions. *Tiling* can also increase parallelism opportunities. Software optimizations for sparse matrix computation, such as OSKI [53] and Sparse BLAS [15], are based on *tiling*. In previous sections, the study of the **`IP`**, **`OP`** and **`Gust`** algorithms highlighted the influence of rank ordering on the sequentiality and data movement, and thus, adding intermediate ranks with *tiling* breaks the regularity of pure **`IP`** or **`OP`** algorithms, requiring additional hardware support (see Section 1.5).

For example, let us consider the A-matrix in Figure 1.5 which is divided into four *tiles* (one level of *tiling*). For a SpMV, two new ranks appear for a total of four ranks, as shown in Algorithm 1.6: M_1 and K_1 allow *tile* selection and M_2 and K_2 allow data access within a *tile*. Thus, we could decide to apply an **`OP`** on *tiles* and conversely an **`IP`** within *tiles*, mixing-up the pros and cons of these two algorithms at different rank levels. In the case of the example in Figure 1.5, the upper-left A-*tile* (*red*) is evaluated first, sequentially updating the upper *tile* of the c-vector in an **`IP`** manner. The lower-left A-*tile* (*yellow*) is then computed the same way. At this point, *intersections* have been done with only the upper b-*tile*, and c has been updated sequentially (first *PO*). The two next A-*tiles* (*green*, then *blue*) will compute *intersections* with the lower b-*tile* and update the entire c-vector sequentially, showing the **`OP`** characteristic of having *POs* to *merge*, two in this case.

DEFINITION

Tiling is a method for partitioning matrices in non-overlapping sub-matrices, called ***tiles***.

NOTE

Note that in case of *tiling*, the A-matrix is no longer read sequentially, which can be problematic if it is stored in a classical sparse format like CSR that is specifically made for *row-wise* accesses. Custom sparse formats adapted for *tiling* can be used [27], but this requires pre-

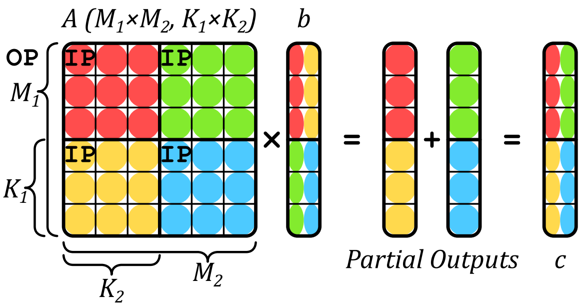


Figure 1.5: An Example of *Tiling* on SpMV

Algorithm 1.6: Matrix-Vector Algorithm with Custom *Tiling* of Figure 1.5

for $k_1 \in [\![0, K_1 - 1]\!]$ **do**
  for $i_1 \in [\![0, M_1 - 1]\!]$ **do**
    for $i_2 \in [\![0, M_2 - 1]\!]$ **do**
      for $k_2 \in [\![0, K_2 - 1]\!]$ **do**
        $i \leftarrow i_1 \times M_2 + i_2$
        $k \leftarrow k_1 \times K_2 + k_2$
        $c_i \mathrel{+}= A_{i,k} \times b_k$

processing to perform the conversion.

CONCLUSION

Tiling methods are widely used to **distribute workloads** across multicore systems, and to **locally reduce the data dimensions**. Complex *tiling* algorithms with many possible rank ordering **inherit the behaviors of IP and OP**, and can be described as a combinations of both at different *tile* levels.

1.5 Hardware Challenges for Sparse Matrix Computation

IN the previous section we presented a brief background on matrix multiplication without considering the underlying hardware. Although the various sparse formats reduce the memory footprint, they have the disadvantage that indirect accesses are required, which creates data-dependencies and reduces the effectiveness of caches. As we have seen, with all three algorithms, at least some of the accesses are irregular. The lack of spatial locality for these accesses reduces the benefit of having caches [40, 58]. In other words, a full cache-line is loaded, but only a few entries are read or modified. Moreover, the matrices of interest are extremely large and even the non-zero elements, and associated metadata, can not generally fit in the cache. Depending on the algorithm, successive accesses to the same element may be too far apart to benefit from temporal locality in the cache.

1.5.1 Inner-Product Algorithm

In an **IP** algorithm, the A-metadata needs to be read to indirectly access the B-matrix (or b-vector), which may itself be stored in a sparse format. In this case, the algorithm needs to check if the non-zero A-indices are present in B (or b). If an index appears in neither list, there is no need to perform the multiplication (ineffectual operation). More generally, this problem applies to any *fiber* and is called the *intersection* search. As a minimum, this requires reading all the metadata for A and B (or b), and identifying those present in both.

In Algorithm 1.7, we present the pseudo-code for an *Inner-Product* (**IP**) SpMSpM with A and B stored in CSR and CSC formats, respectively. The outer-loop (M, line 1) iterates over the rows of A. The middle-loop (N, line 2) iterates over the columns of B. The inner-loop (line 4) iterates over the non-zero elements of the current row (i) of A, and for each index (k_A of position pos_A in the idx array of A, line 5), a search is performed in the metadata of B (line 6), to see if the corresponding index in B is present. Assuming it is ($isMatch$ is true, line 7), the partial sum is computed and then accumulated locally (acc, line 8). This example highlights the need to efficiently search for the *intersection* of indices in the metadata of the inputs.

Different algorithms can be used to address the *intersection* search, depending on whether the elements are sorted. When the elements are sorted, the *intersection* becomes a *merge* and generally requires at most $2 \times nnz - 1$ comparisons (in some cases the number of comparisons can be reduced [5]). In Algorithm 1.7, we use sorted sparse formats (CSR and CSC). Thus, the naïve *intersection* search proposed (line 10) does at most one traversal of each the current row of A (i) and the current column of B (j). Note that computing an *intersection* in software induces the use of conditional loops and branches, which can be costly for classical CPU branch predictors. In case the matrix B (or vector b) is stored in a dense format, the *intersection* search is simplified, since all indices are implicitly present, but the accesses to B (or b) remain indirect. As we see in Chapter 2, certain accelerators provide support for performing the *intersection* operation directly in hardware.

Algorithm 1.7: **IP** SpMSpM Performing *Intersection* Searches with CSR and CSC Formats

Data: $A(M, K)$ in CSR: $A.val, A.idx, A.ptr$
Data: $B(K, N)$ in CSC: $B.val, B.idx, B.ptr$
Data: $C(M, N)$ in dense format (for readability)
Result: $C = A \times B$

1 **for** $i \in [\![0, M-1]\!]$ **do** `// Iterations over the rows of A`
2   **for** $j \in [\![0, N-1]\!]$ **do** `// Iterations over the columns of B`
3         $acc \leftarrow 0$; $pos_B \leftarrow B.ptr_j$;
4   **for** $pos_A \in [\![A.ptr_i, A.ptr_{i+1} - 1]\!]$ **do**
            `// Iterations over the non-zero elements of the i-th row of A`
5             $k_A \leftarrow A.idx_{pos_A}$;
            `// Column-index of the (i, posA) non-zero element of A`
6             $(isMatch, pos_B) \leftarrow IntersectionSearch(k_A, j, pos_B, B.idx, B.ptr)$;
            `// Return true and the matching posB if idx exists in the j-th column of B`
7   **if** $isMatch$ **then** `// If indices did match`
8                 $acc \leftarrow acc + A.val_{pos_A} \times B.val_{pos_B}$;
                `// Local partial sum accumulation`
9         $C_{i,j} \leftarrow acc$; `// Write in memory of the resulting (i, j) element of C`

10 **function** $IntersectionSearch(k_A, j, pos_B, B.idx, B.ptr)$ **is**
    `// Naive intersection search allowing the i-th row of A and the j-th column of B to be traversed only once at iteration (i, j)`
11   **while** $B.idx_{pos_B} < k_A$ ***and*** $pos_B < B.ptr_{j+1}$ **do**
12         $pos_B \leftarrow pos_B + 1$;
13   **if** $B.idx_{pos_B} = k_A$ **then** `// Returns true if indices from A and B match`
14   **return** $(true, pos_B)$;
15   **return** $(false, pos_B)$;

DEFINITION

The ***intersection* search** consists in founding non-zero elements of two sparse arrays that have matching indices. This search is costly in software on classical CPUs, thus being key to improve performance of **IP** algorithm.

1.5.2 Outer-Product Algorithm

In an **OP** algorithm, the inputs A and B (or b) are accessed sequentially, but the updates of the output C (or c) require indirect accesses. The pattern of the accesses, while the output is generated, is irregular and determined by the structure of the inputs. Furthermore, if these "random" updates provoke cache misses where only one word within a cache-line is modified, the bandwidth to external memory is poorly utilized. The **OP** algorithm generates several *partial outputs* (*POs*) that must be accumulated to obtain the final output. We can identify two strategies: *immediate update* and *deferred update*. In the former, the final output is immediately updated as each partial sum is generated. If the output is stored using a sparse format, either the index being updated does not yet exist, and the new partial sum must be inserted. Or, if it exists, a lookup must be done to locate it, the update performed and the result written back. The problem associated with combining the *POs* is called *reduction*.

In Algorithm 1.8, we present two pseudo-code for an *Outer-Product* (**OP**) SpMSpM using *immediate update* (from line 1) and *deferred update* (from line 7) with A and B stored in CSC and CSR formats, respectively. The outer-loop (K, line 1 or 7) iterates over the columns of A and the rows of B (common rank). The middle-loop (line 2 or 9) iterates over the non-zero elements of the current column (k) of A. The inner-loop (line 4 or 11) iterates over the non-zero elements of the current row (k) of B. Because we only iterate over the non-zero values, there is no need for an *intersection* search. The partial sum is then computed (line 6 or 15). Finally, in the *immediate update* version, the partial sum must be accumulated to C (line 6). These accesses are irregular, making it necessary to lookup the value in C, update it, and write it back. This is a *reduction* operation, performed on irregular data.

To avoid the complexities of these "random" *reductions* with *immediate updates*, the *deferred update* version postpones the accumulation and simply stores the partial sums in memory. One such technique is called *output batching* [33]. With this technique, arrays containing the indices and the partial sum are streamed sequentially to memory (PO_k, lines 13-15). Then, during a second pass (line 17), they are read back and the accumulations are performed, an approach which results in sequential memory accesses during both the first and second passes, with the downside that the volume of data transferred to memory is increased. As we see in Chapter 2, hardware accelerators may combine the *immediate* and *deferred update* techniques to optimize memory bandwidth.

DEFINITION

"Random" *reductions* occur when updating data from memory at irregular addresses, which happen on *immediate update* **OP**. With *deferred update* **OP**, the *reductions* are sequential but large sets of data need to be buffered in memory. Optimizing *reduction* operations is key to improving the performance on **OP** algorithm.

1.5.3 Summary

In Table 1.2, we summarize the advantages and hardware challenges for each algorithm, including **Gust** which presents a trade-off between **IP** and **OP**. The potential for parallelism and data-reuse varies across algorithms, **IP** being the most limited unless *tiling* methods are used. With **OP**, *POs* can easily be generated across a multicore system but at the cost of transferring a large volume of data to memory. With *immediate updates*, **OP** transfers less data but all accesses on the output are "random" *reductions*.

Algorithm 1.8: OP SpMSpM Performing *Reduction* with CSR and CSC Formats

Data: $A(M, K)$ in CSR: $A.val, A.idx, A.ptr$
Data: $B(K, N)$ in CSC: $B.val, B.idx, B.ptr$
Data: $PO_k(M, N)$ for all $k \in [\![0, K-1]\!]$ in COO: $PO_k.val, PO_k.row, PO_k.col$
Data: $C(M, N)$ in dense format (for readability)
Result: $C = A \times B$

// *
// * * * 'Immediate updates' version *
for $k \in [\![0, K-1]\!]$ **do** // Iterations over the columns of A and the rows of B
  for $pos_A \in [\![A.ptr_k, A.ptr_{k+1}]\!]$ **do**
    // Iterations over the non-zero elements of the k-th column of A
    $i_A \leftarrow A.idx_{pos_A}$; // Row-index of the ($pos_A$, k) non-zero element of A
    for $pos_B \in [\![B.ptr_k, B.ptr_{k+1}]\!]$ **do**
      // Iterations over the non-zero elements of the k-th row of B
      $j_B \leftarrow B.idx_{pos_B}$;
      // Column-index of the (k, pos_B) non-zero element of B
      $C_{i_A,j_B} \leftarrow C_{i_A,j_B} + A.val_{pos_A} \times B.val_{pos_B}$;
      // Reduction (to memory) of the partial sum into the (i_A, j_B) element of C (immediate update)

// *
// * * * 'Deferred updates' version *
// Multiply phase
for $k \in [\![0, K-1]\!]$ **do** // Iterations over the columns of A and the rows of B
  $pos_{PO} \leftarrow 0$;
  for $pos_A \in [\![A.ptr_k, A.ptr_{k+1}]\!]$ **do**
    // Iterations over the non-zero elements of the k-th column of A
    $i_A \leftarrow A.idx_{pos_A}$; // Row-index of the ($pos_A$, k) non-zero element of A
    for $pos_B \in [\![B.ptr_k, B.ptr_{k+1}]\!]$ **do**
      // Iterations over the non-zero elements of the k-th row of B
      $j_B \leftarrow B.idx_{pos_B}$;
      // Column-index of the (k, pos_B) non-zero element of B
      $PO_k.row_{pos_{PO}} \leftarrow i_A$;
      $PO_k.col_{pos_{PO}} \leftarrow j_B$;
      $PO_k.val_{pos_{PO}} \leftarrow A.val_{pos_A} \times B.val_{pos_B}$;
      // The k-th partial sum of the (i_A, j_B) element of C is stored in memory in the k-th partial output
      $pos_{PO} \leftarrow pos_{PO} + 1$;
  // Each partial output is generated sorted, row-wise
// Merge phase
for $k \in [\![0, K-2]\!]$ **do** // Naive serial merge of the K partial outputs
  $PO_{k+1} \leftarrow \mathit{2DListsMerge}(PO_k, PO_{k+1})$;
// PO_{K-1} is C stored in COO format
for $pos \in [\![0, length(PO_{K-1}) - 1]\!]$ **do**
  // Update C (dense) from the fully merged partial outputs
  $i \leftarrow PO_{K-1}.row_{pos}$;
  $j \leftarrow PO_{K-1}.col_{pos}$;
  $C_{i,j} \leftarrow PO_{K-1}.val_{pos}$; // Write in memory of the (i, j) element of C

Table 1.2: Comparison of Advantages and Challenges with Matrix Multiplication Algorithms

		Inner-Product (`IP`)	***Outer-Product*** (`OP`)	***Gustavson*** (`Gust`)
Access Count	**MV**	***A*:** once ***b*:** $nnz(A)$ ($\sim M\times$) ***c*:** once	***A*:** once ***b*:** once ***c*:** $nnz(A)$ ($\sim K\times$)	*N/A*
	MM	**A:** $N\times$ **B:** $M\times$ **C:** once	**A:** once **B:** $M \times d(A)\times$ **C:** $nnz(A) \times \overline{nnz_r}(B)$ ($\sim K\times$)	**A:** once **B:** $M \times d(A)\times$ **C:** $nnz(A) \times \overline{nnz_r}(B)$
Sequential Accesses		Accesses to A and C	Accesses to A and B	Accesses to A and C Accesses to B-rows
Input Format		Accesses to A and B benefit from alternate formats (CSR and CSC)		Allows consistent format for A and B
Parallelism Opportunity		Possible with *tiling*	Generation of *POs* over common rank of A and B	Reading A-rows and updating C-rows
Reuse Opportunity		Potentially B, but limited by its size	An A-column or B-row while computing *POs*	A set of B-rows, based on non-zero elements of A-rows
Challenges		*Intersection* search between non-zero elements in A and B	Storage for *POs* or in-place *merge* and *reduce*	*Merge* and *reduce* of *PO*-rows

The issues presented with **`IP`** and **`OP`** can be generalized to two concepts: *gather* and *scatter*. The *gather* problem exists when the inputs are indirectly accessed, and need to be read from memory using metadata to present the processor with the correct terms for computing the partial sums. Typically, with **`IP`**, the *gather* problem is more difficult. The *scatter* problem exists when the order in which the output matrix/vector is generated is not sequential and is typically associated with the **`OP`** algorithm. In the case of *tiling* or with **`Gust`**, both *gather* and *scatter* must be performed creating a need for efficient *intersection* searches and random *reductions*.

CONCLUSION

There are **two main challenges** to address when computing sparse matrices. When reading irregular data from memory, addressing the ***gather*** problem is key. It consists of processing efficient ***intersection* searches**. When updating irregular data from memory, addressing the ***scatter*** problem is key. It consists of **managing *POs*** in memory or processing efficient **"random" *reductions***. These two challenges are tied to **`IP`** an **`OP`**, respectively, and are present in any higher rank algorithms for sparse matrix computing.

1.5.4 SpMSpM Algorithm Analysis

To better understand how the algorithms impact the number and types of memory accesses, we developed, and presented in Figure 1.6, a model based on memory accesses counts for SpMSpM. We identify five memory access patterns: (1) those where a matrix is read only once, thus there is no opportunity for reuse, (2) where the accesses are sequential, but a *fiber* is read multiple times, (3) where *fibers* are read consecutively, but the accesses within a *fiber* are "random", (4) where the *fibers* are accessed "randomly", but within a *fiber* the accesses are sequential, and (5) where the matrix elements are accessed multiple times, and with a "random" access pattern.

For each of the three algorithms, we counted the number of read and write accesses, based on the density of the input matrices, and the local memory storage available ("0D", "1D" and "2D"), as presented in Table 1.3. We refer to a system with no local storage or cache as '0 Dimensional' ("0D"), a system which has local storage that is sufficient to hold a single row or column as '1 Dimensional' ("1D"), and a system which can cache an entire matrix as '2 Dimensional' ("2D"). In Figure 1.6, we plotted the number of memory accesses for SpMSpM, in a logarithmic scale, based on a square matrix with $M = 10,000$. Indeed, the number of non-zero elements in C depends on the structure of the input matrices and in this model, we have assumed the non-zero entries in the input matrices are uniformly, randomly distributed. This corresponds to a *worst case* for sparse matrices (no spatial and temporal locality). If A or B are banded matrices, for example, there would be fewer non-zero elements in C. In each graph, we have separated the accesses into A (*bottom*), B (*middle*) and C (*top*). The color code shows the access pattern, based on the five categories described above.

Table 1.3: Memory Access Count for Each Matrices of `IP`, `OP` and `Gust` SpMSpM

	Inner-Product (`IP`)	***Outer-Product*** (`OP`)	***Gustavson*** (`Gust`)
A(M, K)	**0D**: $N \times nnz(A)$ **1D-2D**: $nnz(A)$	**0D-1D-2D**: $nnz(A)$	**0D-1D-2D**: $nnz(A)$
B(K, N)	**0D-1D**: $M \times nnz(B)$ **2D**: $nnz(B)$	**0D**: $M \times d(A) \times nnz(B)$ **1D-2D**: $nnz(B)$	**0D-1D**: $M \times d(A) \times nnz(B)$ **2D**: $nnz(B)$
C(M, N)	**0D-1D-2D**: $nnz(C)$ (max: $M \times N$)[1]	**0D-1D**: $2 \times M \times d(A) \times nnz(B)$ **2D**: $nnz(C)$ (max: $M \times N$)[1]	**0D**: $2 \times M \times d(A) \times nnz(B)$ **1D-2D**: $nnz(C)$ (max: $M \times N$)[1]

[1] We can not easily quantify $nnz(C)$ knowing $nnz(A)$ and $nnz(B)$. We thus use a statistic approximation.

In the top-row of Figure 1.6 (graphs ①, ② and ③), we start by considering a naïve system which has no local storage ("0D"), thus all data comes from main memory. From the graphs in this row, we clearly see that `IP` requires more accesses at low density, as the A-matrix must be read N times (graph ①). Whereas for `OP` and `Gust`, the A-matrix is read only once (thus the *grey* color). For `OP` and `Gust` the total number of accesses is the same, however, with `OP` (graph ②), the accesses to C are "random" (*red*). With `Gust` (graph ③), the accesses to B are sequential within the rows (*orange*) and the C-rows are generated sequentially (*yellow*), thus `Gust` avoids having any fully "random" accesses (*red*).

Since main memory accesses are the principal bottleneck, systems integrate local storage (such as buffers or caches). Thus, we consider the case of a system with sufficient local memory to store one full row, per matrix (equivalent to few sparse rows, "1D"). We assume this local storage is perfect (no misses) and we re-calculate the number of remaining main memory accesses, as shown in the middle-row of Figure 1.6 (graphs ④, ⑤ and ⑥). The matrices are assumed to be initially stored in main memory, thus they must be accessed at least once. As can be seen, for `IP` (graph ④), the number of A-accesses is reduced, because rows are reused, thus the total number of A-accesses becomes equal to that in `OP` and `Gust`. For `OP` (graph ⑤), the single row buffer greatly reduces the number of B-accesses, however, it does nothing for C (despite the appearance, due to the logarithmic scale). For `Gust` (graph ⑥), the single row buffer is sufficient to cache the *reduction* operations in C, and thus, it now has fewer overall accesses than `OP`. It is also interesting to note that `OP` and `Gust` scale better than `IP` with lower densities, as seen by the shallower slope.

Finally, we have considered the extreme case where an system has a buffer of sufficient size to store an entire matrix ("2D"), which is shown in the bottom-row of Figure 1.6 (graphs ⑦,

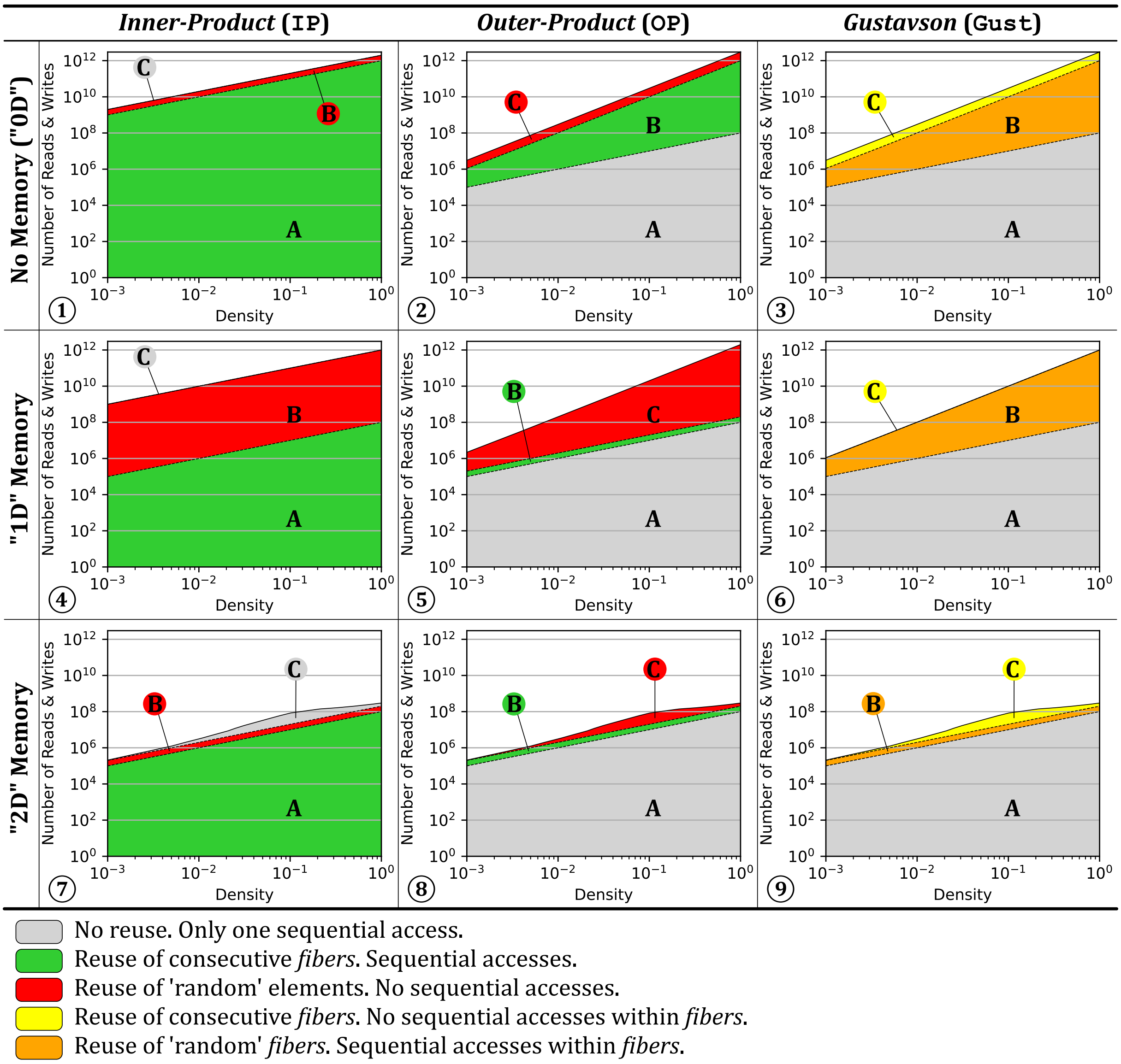


Figure 1.6: Model of Memory Access Counts Depending on Algorithm and Local Memory Size

⑧ and ⑨). Although this case is not practical for large matrices, through the proper use of *tiling* to locally reduce the dimensions of a sub-matrix, this case can be achieved at the level of a *tile*. In this case, the total number of external accesses is equal for the three algorithms, as each matrix is accessed only once from main memory. For **`IP`** (graph ⑦), it is interesting to note that such a large buffer is needed to locally address the *intersection* search in B. Similarly for **`OP`** (graph ⑧), a large storage is required to address the *reductions* in C. For **`Gust`**, in fact, it is not necessary to buffer the entire B-matrix. Indeed, buffering a few rows (corresponding to $nnz_r(A)$) is sufficient to achieve the number of accesses shown in graph ⑨. This corresponds to an intermediate design point (multiple row buffers for B) not explicitly shown in the figure (between graphs ⑥ and ⑨).

When viewed overall, Figure 1.6 highlights the benefits of **`Gust`**. We clearly see that **`Gust`** does not require any fully "random" accesses (*red*). With a single row buffer, it is possible to address the *reductions* in C. And, with a limited number of row buffers, the number of external memory accesses for B can be reduced. Based on this analysis, it is clear that, among the three base algorithms, **`Gust`** is the best candidate for hardware implementation of SpMSpM.

NOTE

For SpMV, such an analysis does not show significant differences between **`IP`** and **`OP`**, as both algorithms require similar number of memory accesses. However, **`OP`** shows more potential for parallelism and data-reuse as seen in the previous section (1.5.3), thus being preferred for SpMV hardware accelerators.

CONCLUSION

Based on a high level analysis of the memory accesses on SpMSpM, ***Gustavson*'s algorithm** appears as the **best trade-off** to minimize memory traffic with reasonable amount of local storage. The second best option is **`OP`**, which requires fewer memory accesses than **`IP`** when the matrices have a low density.

1.6 Conclusion

In this chapter, we established the foundation for understanding the relationship between applications working with irregular data and the underlying challenges in hardware. Irregular accesses are common in sparse computing and cause reduced cache efficiency, limited parallelism, and higher memory latency exposure. We analysed the most representative algorithms for sparse matrix multiplication kernels, and identified the operation that provoke irregular accesses. *Intersections* and *reductions* on irregular data with classical systems are costly, thus they represent the central challenge for efficient computation on sparse data. To improve the performance of these operations, specialized hardware has been proposed as will be seen in the next chapter.

TAKEAWAYS

Sparse matrices: large, highly sparse, structured (such as banded matrices)
Sparse formats: diverse, reduce memory footprint

- COO, the simplest
- CSR and CSC, the mostly used
- BaCSC, our own sparse format for banded matrices

Sparse kernels: sparse MV and MM multiplications (such as SpMV and SpMSpM), generate indirect accesses
Algorithms: *Inner-Product* (`IP`), *Outer-Product* (`OP`) and *Gustavson* (`Gust`), extended with *tiling* methods

- Best potential to reduce memory transfers: `Gust`, followed by `OP` at low density

Problem: irregular accesses, generate high memory transaction latencies
Challenges:

- *gather*: *intersection* searches
- *scatter*: “random” *reductions*, or storing and *merging partial outputs* (*POs*)

Chapter 2

State-of-the-Art

Contents

2.1 Introduction

TODO

Chapter 3

High Level Architectural Presentation of SpDCache

Contents

3.1 Introduction

In the previous chapters, we analyzed the causes and effects of irregular accesses in sparse computations and identified the *Outer-Product* (**OP**) algorithm of Sparse Matrix-Vector multiplication (SpMV) ($A \times b = c$) as a key case. The **OP** algorithm generates irregular updates on the output vector c, as each non-zero element of the input matrix A contributes to distinct, and often distant, memory locations. These "random" *reductions* are a major challenge for cache-based architectures.

Conventional cache hierarchies are designed for regular, predictable access streams that exploit spatial and temporal locality. In contrast, **OP** SpMV exhibits poor locality, leading to low cache utilization, frequent evictions, and inefficient bandwidth use.

Previous works, such as PHI [41], SortCache [48], and ASA [61], have proposed *reduction*-aware cache mechanisms to alleviate irregularity by improving accumulation and storage efficiency. However, these solutions do not exploit the structural properties of the processed matrices, and therefore miss optimization opportunities when partial regularity is present. This observation motivates our focus on banded sparse matrices, which combine overall sparsity with localized dense regions near the diagonal. Such matrices provide a controlled environment to study how structure and locality can be leveraged to enhance cache performance.

This chapter begins by recalling the main concepts of generic data-caches, introducing the terminology and mechanisms required to discuss caching behavior under irregular access conditions. It then presents a *reduction*-aware cache architecture, designed to efficiently support “random” *reductions* in **OP** SpMV. The proposed design, SpDCache (presented in Paper II), extends traditional caching models with region-based organization and native *reduction* support. Finally, we evaluate its performance through high-level modeling on banded and non-banded matrices, highlighting the influence of data regularity on cache efficiency.

3.2 Generic Data-Caches

MODERN processors and accelerators integrate data-caches to bridge the latency and bandwidth gap between computation units and main memory. A cache is a small, fast memory that stores recently accessed data to anticipate future reuse. Its efficiency relies on two main forms of locality:

- temporal locality, where recently accessed data is likely to be reused soon, and
- spatial locality, where data stored near recently accessed addresses is likely to be used next.

By exploiting these properties, caches reduce the average access latency and increase the overall bandwidth efficiency.

A cache is composed of several cache-lines, each grouping contiguous memory words. The cache-line is the minimum granularity of data movement between memory levels. When an access occurs, the cache controller searches for the requested address among the currently stored cache-lines. If the address is found, a hit occurs and the data is served directly to the core. Otherwise, a miss occurs, triggering a memory fetch from a lower cache level or from main memory. If necessary, an existing cache-line is evicted to make room for the new data. Each cache-line is associated with *tags* and validity bits that record its address and coherence state.

Several mapping policies exist. In a direct-mapped cache, each memory block maps to a single cache-line, which is simple but prone to conflicts. A *n-way set*-associative cache allows each block to be placed in one of n possible cache-lines within a *set*, improving hit rate at moderate cost. A *fully associative* cache removes these constraints, as any block may occupy any cache-line, but requires more complex *tag* searches. When multiple candidates exist, a replacement policy such as *Least Recently Used* (*LRU*) or *Random* decides which entry is evicted.

Write policies also impact performance. With a *write-through* policy, every store operation updates both the cache and main memory, maintaining consistency but increasing traffic. A *write-back* policy delays the update until eviction, reducing bandwidth usage at the cost of additional state bits (dirty flags) to maintain consistency. For workloads dominated by sequential accesses, these mechanisms are highly effective. However, irregular accesses tend to degrade cache efficiency by increasing miss rates and provoking frequent evictions.

Modern processors implement hierarchical cache organizations to balance latency, capacity, and throughput. A typical CPU integrates small L1 caches close to each core, larger L2 caches, and a high-capacity L3 cache shared at the chip level. Accelerators and domain-specific architectures often employ local data storages or scratchpad memories serving the same purpose, providing low-latency access to frequently reused data (see Chapter 2).

The generic cache assumes regular read and write operations with predictable locality. Yet workloads operating on sparse or *reduction*-intensive data violate this assumption. In particular, "random" *reductions* are poorly supported, as conventional caches lack native mechanisms to merge updates efficiently or to retain relevant data in limited storage.

In the next sections, we introduce cache architectures supporting *reduction* operations, specifically designed to handle irregular data. These architectures generalize the concept of a data-cache to include *reduction*-aware mechanisms, ensuring correctness while improving bandwidth efficiency.

In the following sections, we introduce cache architectures with native *reduction* support, specifically designed to handle irregular data. These architectures extend the classical cache model with *reduction*-aware mechanisms, improving bandwidth efficiency under irregular access patterns.

CONCLUSION

Generic data-caches efficiently exploit regular access patterns but **fail to address the irregularity** and accumulation requirements of sparse data computing. These limitations motivate the introduction of ***reduction*-aware cache architectures**, capable of handling irregular data.

3.3 Reduction Cache

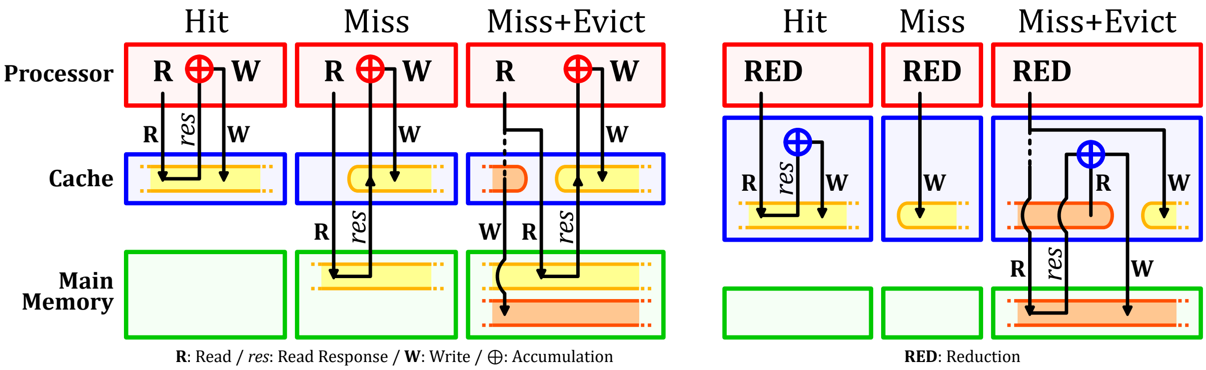
PERFORMING a *reduction* in hardware requires a read from memory, followed by an accumulation and a write back. This operation is denoted $RED(key, value)$ [48] with the *key* being the position in the output vector c. In Figure 3.1a, we show three cases that occur in a generic data-cache when a *reduction* is performed. If the vector entry (c_{key}) hits in the cache, the *update* is performed by the processor (in *red*) with no external memory access (on the left in the figure). When there is a miss, the operation is blocked until the read from main memory completes (in the middle of the figure). Potentially, the miss will also trigger an eviction, which exacerbates the blocking condition, because a cache-line must first be written back, before the read from main memory can be performed (on the right in the figure). In scientific applications, the matrices are extremely large and only a small portion of the vector c can reside in the cache, thus the *miss+evict* case arises frequently.

A *reduction cache*, such as [48] or [41], is a data-cache which accepts *reduction* instructions (*RED*) from the processor and performs accumulations locally (near-memory). In the case of a hit, see Figure 3.1b, the *reduction* takes place in the cache (in *blue*, on the left in the figure). For a miss, the cache allocates a new line with zeros, and simply inserts the value (in the middle of the figure). In this case, the cache-line is inconsistent with main memory. Only when the line needs to be evicted, is a read performed (on the right in the figure). In the cache, the values in the line are accumulated with the main memory data, and the result is written back.

As seen in the figure, a *reduction cache* reduces the number of main memory accesses in the case of a miss. Instead of having the classic ascendant policy where data is stored in the cache to be pulled-up again by the processor latter-on, the *reduction cache* has a descendant policy: data is written/updated in the cache to be pushed-down to memory latter-on. Instead of two memory transactions that can stall the processor, the *reduction cache* only needs a single “fire-and-forget” instruction, meaning it can be issued without waiting for an answer, thus avoiding stall issues. At the end of the algorithm, the *reduction cache* must be flushed.

DEFINITION

A ***reduction cache*** is a data-cache containing an arithmetic unit and a dedicated support for *reduction* operations, via single “fire-and-forget” *RED* instructions.



(a) Generic Cache (Write-Back) **(b)** Reduction Cache

Figure 3.1: Read (R) and Write (W) Transactions of a *Reduction* Operation in Generic and *Reduction* Caches

CONCLUSION

In order to process efficient *reduction* operations, a ***reduction cache*** can be used to **avoid unnecessary memory transactions and stalls**, by offloading the accumulation near-memory, and updating the main memory only at eviction.

3.4 Outer-Product SpMV with a Reduction Cache

A widely used sparse format to compress the non-zero elements in a matrix A is CSC [46], where A is a sparse matrix of dimensions (M, N) with a number of non-zero elements nnz. As seen in Chapter 1, it consists of three tables (val, idx, ptr) used to store non-zero values, row indices and column positions. When performing an **OP** SpMV ($A \times b = c$) with this format (Algorithm 3.1), we see that a *RED* is needed to accumulate the output vector c (line 6). Moreover, A is read sequentially while c is updated "randomly". In fact, the *key* is determined via an indirection with an access to idx.

Algorithm 3.1: *Outer-Product* SpMV – Single Threaded

1 **for** $n \in [\![0, N-1]\!]$ **do** // Loop over the columns of A
2 **for** $pos \in [\![ptr_n, ptr_{n+1} - 1]\!]$ **do**
3   $key \leftarrow idx_{pos}$; // Row index
4   $val \leftarrow val_{pos} \times b_n$; // val = $A_{key,n} \times b_n$
5   $RED(key, val)$; // c_{key} += val

In memory, A is stored as three dense tables, but conceptually, as we perform the **OP** multiplication, the matrix is traversed from top to bottom, then left to right. A row in the matrix corresponds to a *key*, or equivalently, a position in the output vector c. In Figure 3.2, we show an example of a sparse matrix with a few non-zero values (① - ⑥). As we traverse the matrix, we first perform a *reduction* for the value on row key_{10}, then on row key_{20} and then key_{30}. Each of these results in a *RED* on c. As we continue, we then hit another value ⑤ on the row key_{10} which must be *reduced* with an existing entry ① (as shown on the left of Figure 3.2, *Conceptual View*).

Let us consider how c is stored in a two-*way set*-associative cache with 32B lines. We assume that all the *keys* map to set_1. When ① is updated, there is a miss and $line_1$ is allocated

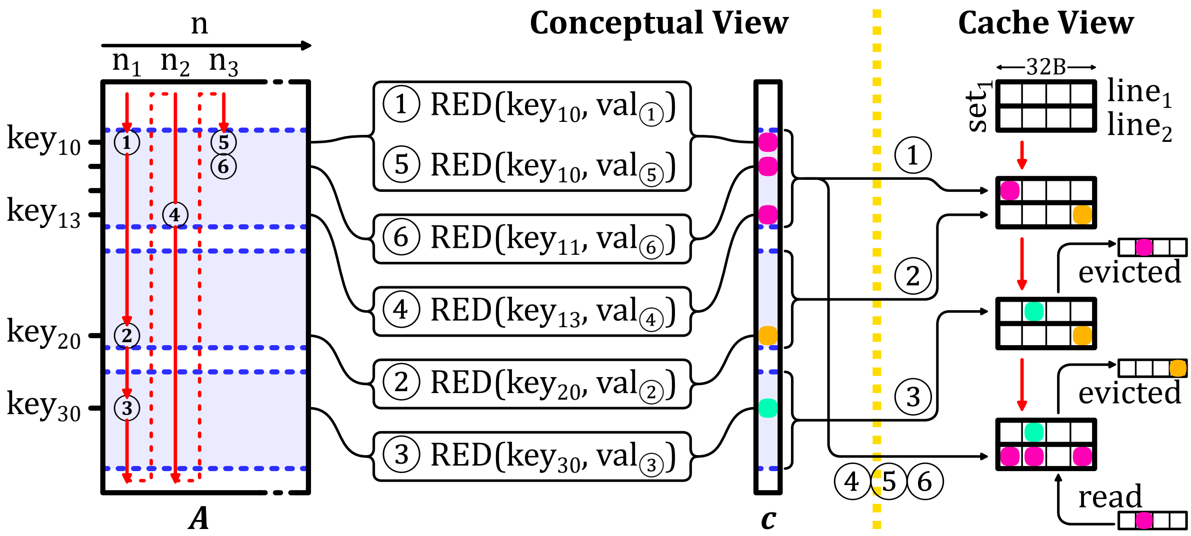


Figure 3.2: Traversal of Non-Zero Elements of Input Matrix for OP SpMV (left) and Impact on Cache (right)

for key_{10}. Similarly with ②, $line_2$ is allocated for key_{20}. When key_{30} comes along, an eviction must be performed. We assume $line_1$, holding key_{10}, is evicted. We then come to ④ which has key_{13}, adjacent in memory to key_{10}, and thus stored in the same line. This line must be brought back to process ④, ⑤ and ⑥.

This example illustrates how a conventional cache can be inefficient. First, we note the line holding key_{10} was evicted, although it was frequently accessed after ①. This eviction was triggered by key_{30}, which was only accessed once. Furthermore, for key_{20} and key_{30}, a full line was loaded, only to modify a single entry. A “smarter” cache would have maintained the line with key_{10} to avoid an unnecessary main memory access. Furthermore, this “smarter” cache would not have allocated full lines for key_{20} and key_{30} to only modify a single word.

The globally low local and temporal localities avoid caches, and even *reduction caches*, to take advantage of any reuse opportunity. However, sparse matrices have various types of structure which can have reuse potential on some parts of the matrix. In the example in Figure 3.2, the first rows (on key_{10} to key_{13}) are denser than the rest of the matrix. A “smarter” cache would prefer to focus on these rows, keeping the corresponding cache-line and accumulating every elements before a final eviction. On the contrary, the sparser rows of the matrix would not stayed long in the “smarter” cache, having low priority to avoid thrashing data from denser regions.

We focus on the case of banded matrices which have a denser region around the diagonal. For a *reduction cache* to work efficiently with such matrices, the dense band should be prioritized, and the isolated elements out of the band should be processed separately to avoid disturbance. This is the main idea of SpDCache (Paper II), which is a *reduction cache* with dedicated memory regions for the in-band and out-of-band of a matrix. We describe its behavior int the following sections.

CONCLUSION

When computing an **OP SpMV**, a **conventional *reduction cache* is not enough** for the available cache memory to be efficiently utilized. Compared to the size of a cache, matrices have too large dimensions for *keys* to be reused from one column iteration to an other, leading to mostly empty cache-lines being prematurely evicted.

We propose **SpDCache**, a *reduction cache* with dedicated memory regions, **leveraging matrix structure** with denser and sparser regions, such as **banded matrices**.

3.5 Architecture of SpDCache

SPDCACHE is developed to replace a L1 generic data-cache. Thus, it has to function as a generic cache when receiving classical memory instructions from the processor, such as reads and writes. Indeed, when processing an **OP** SpMV, the sequential reads to the matrix A and the vector b should not bypass the cache hierarchy which takes advantage of spatial locality. Only, the output vector c suffers from "random" *reductions* that should be processed with a region-wise *reduction cache*. SpDCache can be extended on a multilevel memory hierarchy, as well as a multicore architecture (see Chapter 7). However, our study (up to Chapter 6) focus on a single level cache hierarchy with a single core, to focus on the SpDCache behavior.

As shown in Figure 3.3, SpDCache is divided into three regions. The first, *Generic Region Cache* (*GRC*), is for all memory transactions unrelated to the output vector c, including reads of A and b. It is a classic, *set*-associative cache and not our focus, but is required for the processor to operate normally. The other two regions, called *Dense Region Cache* (*DRC*) and *Sparse Region Table* (*SRT*), are able to perform *reductions* on c. The *DRC* works as a line-wise, *set*-associative *reduction cache* to hold high density data in the *In-Band Region* of the matrix (*IBR*, *orange*). The difference with the *GRC* is that it is able to perform *reductions* and it is dedicated to the *banded* region, so that spurious accesses do not provoke undesired evictions. Finally, the *SRT* works as an element-wise, fully-associative table whose role is to handle sparse regions, such as the *Out-of-Band Region* in the matrix (*OBR*, *blue*).

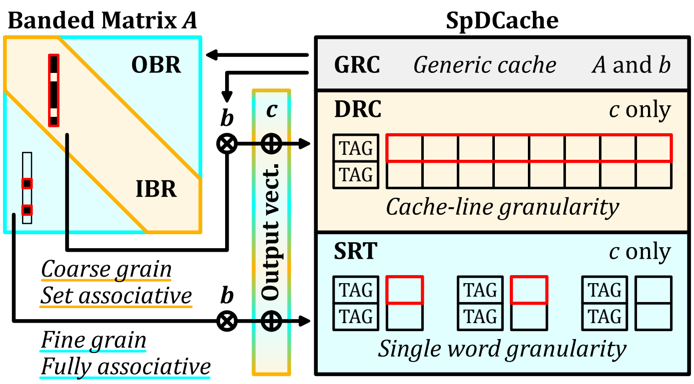


Figure 3.3: Data-Cache Partitioning into Regions (*GRC*, *DRC*, *SRT*)

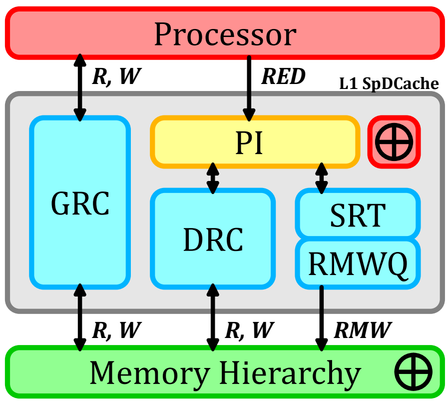


Figure 3.4: Simplified View of the Memory Hierarchy with SpDCache

Figure 3.4 shows a simplified view of a memory hierarchy with SpDCache. The *Processor Interface* (*PI*) takes *reduction* instructions, *RED(key, value, region)*, from the processor and dispatches them to the *DRC* and *SRT*, depending on the *region* argument. The *DRC* communicates with the memory hierarchy using read and write transactions over entire cache-lines, to propagate the local *updates* to the downstream memory. As *SRT updates* are for isolated elements, it does not make sense to bring a full line into the L1 to modify a single word.

The *SRT* thus propagates *updates* by issuing atomic *Read-Modify-Write* transactions (*RMWs*), which must be processed remotely and close to the main memory. Existing protocols such as AXI-4 [1] already support atomic transactions for certain operations (logical, integer add, min, max) and these would need to be extended to include floating point addition. The *RMWs* from the *SRT* are temporarily buffered in a small queue (*RMWQ*), to avoid congestion.

SpDCache is driven by three key design principles. First, we ensure a given *key* is in only one region, to avoid coherency problems. Second, the *DRC* is given priority over the *SRT*, to enable coalescing. Third, we avoid bringing a full line from memory to the L1 to simply update a few entries.

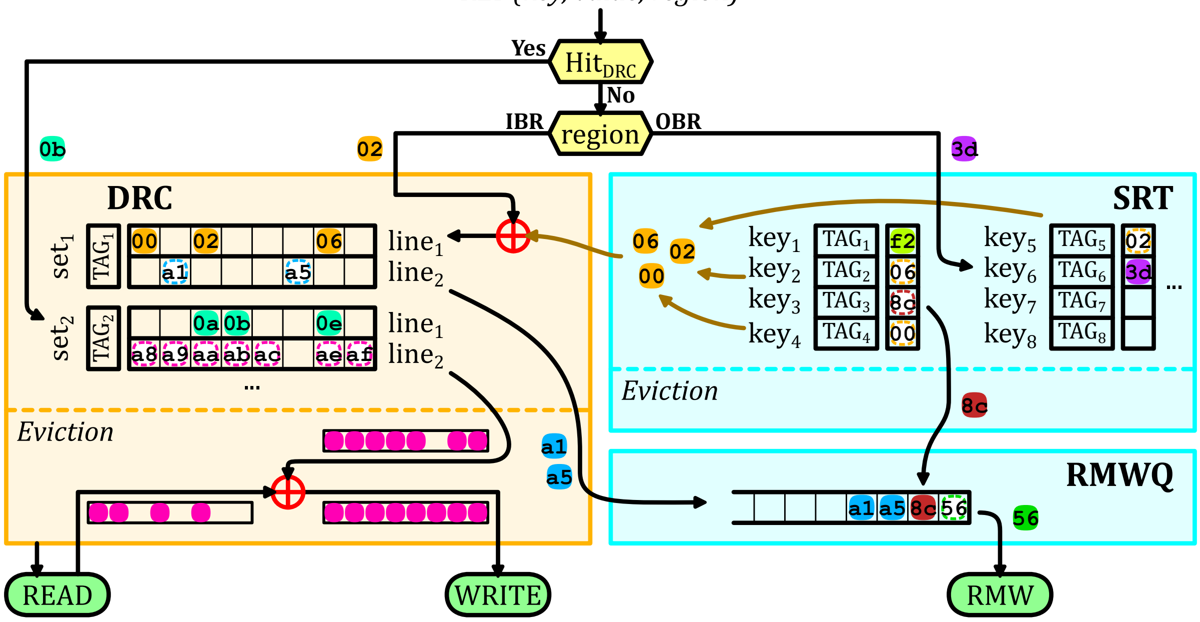


Figure 3.5: Internal Operation of SpDCache with Examples

In Figure 3.5, we describe the operation of SpDCache, via a series of scenarios, each shown in a different color. When a new *reduction RED(key, value, region)* is received from the processor, we test for a hit in the *DRC*, regardless of the *region* argument. If it hits, the value (case `0b` – *turquoise*) can be accumulated in the corresponding cache-line of the *DRC*. We give this priority to the *DRC* as part of the mechanism to avoid duplication.

If there is a miss in the *DRC*, then the target region is determined based on the *region* argument of the *RED* transaction. If it is *IBR*, then it is necessarily a miss case, and the value (`02` – *orange*) must be inserted into a free cache-line of the *DRC*. When we allocate this line, we ensure that it does not create a duplication of existing entries in the *SRT*. If there is duplication, as in our example, these entries (`00`, `02`, `06`) are deleted from the *SRT* and merged into the newly allocated cache-line $(set_1, line_1)$.

If the target region is the *OBR*, then the *reduction* (`3d` – *purple*) is processed in the *SRT*, either allocating a new entry (miss case) or accumulating with an existing entry (hit case).

Now, we describe the eviction scenarios. In the *DRC*, the line to evict is selected $(set_2, line_2$ – *pink*), a read to main memory is performed, the values from the evicted cache-line are accumulated and a write to main memory is issued. If the evicted cache-line only has a few non-zero elements and the *RMWQ* has enough free entries, it is a special case. To avoid bringing a full line from main memory, the non-zero values (`a1` and `a5` – *blue*) are converted to atomic *RMWs* and pushed onto the *RMWQ*. The decision to convert evicted cache-lines to atomic *RMWs* is based on a configurable density threshold. Previous work [41] used a threshold of one non-zero word per eight words in a cache-line. Given our queuing mechanism to handle congestion, we use a more aggressive threshold of three non-zero words per cache-line.

The width of the *band* (w_B), must be selected so that one column of the *band* remains blocked in the *DRC*, otherwise, this region of the cache is always thrashing. This is our approach to selecting w_B, which we detail in the next chapter. Note that this approach selects w_B based on the *DRC* size, independently of the matrix and its structure.

In the *SRT*, an eviction simply consists of moving the element from the table to the *RMWQ* (8c – *brown*). If the *RMWQ* is not full, *SRT* evictions are issued in advance when the *SRT* table is nearly full, using an other configurable threshold sets to half the size of the queue in our experiments. The *RMWQ* issues atomic *RMWs* when it is not empty (5b – *dark green*).

NOTE

SpDCache is not limited to banded matrices. Any sparse matrix with a denser region can benefit from this architecture, by adapting the definition of the *band* accordingly, and making sure that the blocking effect in the *DRC* is preserved.

CONCLUSION

SpDCache is a data-cache splits in **three regions**:

- *GRC*, a **generic** cache region processing reads and writes of A and b,
- *DRC*, a *reduction cache* region dedicated to *reductions* on c when computing the **band** (*IBR*),
- *SRT*, a *reduction cache* region dedicated to *reductions* on c when computing the isolated **out-of-band** elements (*OBR*).

The two last regions use **adapted memory transactions** to avoid transferring unnecessary data.

3.6 High Level Evaluation of SpDCache

WE modelled the behavior of SpDCache, based on the flowchart in Figure 3.5, using a transaction level model in Python, for a single-core, single-thread system executing a OP SpMV kernel (Algorithm 3.1). We do not model timing, as the order of the memory accesses determines the behavior of the cache (hits, misses, evictions), which depends only on the SpMV algorithm. The model checks the final numerical result and reports the main memory traffic associated with the *updates* to the output vector (*Output Memory Traffic*), where we count the bytes transferred (64B for each read and write, and 8B per atomic *RMW*, with our configuration). For each write transaction, it tracks how many words in the line were actually used (*Evicted Cache Line Utilization*) and it tracks how many words in the cache held useful data (*Cache Memory Utilization*). We consider a word in the cache to be useful if it is updated or inserted by the processor. Our goal is to validate the principle of SpDCache using real application matrices from the SuiteSparse Matrix Collection [35].

As baselines, we implemented a fully generic cache (GC) that handle reads and writes for both inputs (A, b) and output (c), following the data movements shown in Figure 3.1a, and a simple *reduction cache* (RedC) that processes *RED* transactions in a dedicated region using the flow shown in Figure 3.1b. In the RedC, 25% of the memory is allocated for the inputs and the other 75% for *REDs* on c. In SpDCache (SpDC), 25% is allocated for the *GRC*, 50% for the *DRC* and 25% for the *SRT*. In the next section, we present more results with varying region sizes, showing that this configuration is a good compromise. All three caches use a *Least Recently Used* (*LRU*) eviction policy, except the *SRT* region of SpDC which uses a *random* policy. All three caches have a total data storage of 32 KiB, and we do not account for the *TAG* storage. The *set*-associative regions have 4 *ways* and 64B cache-lines (8 *double* words per line).

We simulated 48 matrices that have been used by the state-of-the-art accelerators [4, 26, 42, 48, 49, 61, 62], which are listed in Appendix D. We excluded matrices whose output vector c fits entirely in the cache (fewer than 4096 elements), since they do not stress the cache. In Table 3.1, we report the results for all three caches, along with statistical indicators. Most, but not all, of the matrices have a dense band and even some matrices without an obvious band benefit from the blocking effect in the *DRC*.

Table 3.1: Decrease in Memory Traffic and Improvement in Cache Utilization from Python Simulation

	Output Memory Traffic			Evicted Cache Line Utilization			Cache Memory Utilization (%)		
	RedC	**SpDC**		**GC**	**RedC**	**SpDC**	**GC**	**RedC**	**SpDC**
	%/GC	*%/GC*	% RMW	*Avg*	*Avg*	*Avg*	*Avg*	*Avg*	*Avg*
Geometric Mean	89.2	24.5 [a]	40.5	3.60	3.56	7.21 [b]	36.4	45.8	75.2 [c]
Standard Deviation	29.8	20.9	33.8	2.45	2.77	2.08	10.9	32.3	18.6
Minimum	28.5	7.00	0.00	1.06	1.00	4.36	16.6	11.9	30.0
Maximum	172	98.4	100	8.00	8.00	8.00	49.6	97.5	98.9
First Quartile	76.0	11.4	35.9	1.76	1.73	6.03	28.4	22.0	65.5
Third Quartile	115	46.6	91.7	6.60	7.65	8.00	48.6	88.8	93.9
Outperform GC	45.8	100	Ø	Ø	45.8	91.7	Ø	64.6	100
Outperform RedC	Ø	93.8	Ø	Ø	Ø	85.4	Ø	Ø	77.1

[a] *Decrease by 4.1× compared to GC, and 3.6× compared to RedC.*
[b] *Increase by 2.0× compared to GC, and 2.0× compared to RedC.*
[c] *Increase by 2.1× compared to GC, and 1.6× compared to RedC.*

As expected, the simple *reduction cache* (RedC) decreases the main memory traffic to 89.2% of the GC baseline. SpDCache has significantly better performance than RedC, reducing the traffic to 24.5%. SpDCache thus achieves a 4.1× decrease in memory traffic compared to GC, and a 3.6× decrease compared to RedC, on average over the set of matrices. SpDC outperforms GC on all matrices, and RedC on 93.8% of them.

The benefits of SpDCache can be partly explained by the improved utilization of the cache-lines. On average, a cache-line evicted from SpDC has 7.21 modified words, compared to 3.56 and 3.60 with the baselines. This improved utilization is the result of blocking the dense region in the *DRC*, where multiple *reductions* have the time to accumulate in the cache before being evicted. The *SRT* also contributes to this effect, as it captures irregular accesses outside the band, which would otherwise evict useful data from the *DRC*. The overall cache memory utilization is also significantly improved, with 75.2% of the cache holding useful data in SpDC, compared to 45.8% and 36.4% for RedC and GC, respectively.

We need to consider an additional metric, the percentage of traffic sent as atomic *RMW* transactions (% RMW). If this value is too high, it means we have simply shifted the workload to the lower level caches, which inherently have limited compute capability. In mean, 40.5% of the memory traffic of SpDCache is composed of atomic *RMWs*, which is acceptable given the significant overall reduction in memory traffic. However, this ratio varies widely between matrices. There is some matrices where most of the traffic is composed of atomic *RMWs*, meaning that most of their non-zero values are outside the band. For such matrices, which are basically not banded, SpDC still greatly decrease the memory traffic, but better performance could be achieved by choosing a matrix-fitting region partitioning strategy. For example, the matrix *cit-Patents* has no elements in the band, and all non-zero values are scattered under the bottom part of the matrix (an unusual extreme case). While having a memory traffic decreased to 7.2% compared to GC, it could benefit from a better use of the *DRC* by choosing a horizontal band covering the non-zero elements, with correct sizing to leverage the blocking effect. On the other hand, some matrices have very low *RMW* usage, indicating that most non-zero values are within the band and efficiently handled by the *DRC*. In this case, 25% of the available memory was unused. The *SRT* size could be adjusted to gain performance.

CONCLUSION

SpDCache achieves a 4.1× **decrease in memory traffic** on the output vector c compared to a generic cache, over a set of 48 matrices. It also **outperforms a simple *reduction cache*** by 3.6×, showing the importance of having separate dedicated cache regions for the in- and out-of-band of a matrix.

3.7 State-of-the-Art Comparison and Region Sizing

To compare SpDCache with the state-of-the-art, we simulated PHI [41], the existing solution that is closest to our proposal, using the same methodology as in the previous section. PHI is a *reduction*-aware cache that uses a single region to process *RED* transactions (see Section **??** from previous chapter). PHI works as if our three regions, *GRC*, *DRC* and *SRT*, where merged together, using different technics to handle *reductions*, load and store operations in the same cache address space. It also handles isolated elements, but do not consider the matrix structure. Conversely to SpDC, PHI relies on the use of the *deferred update* version of the **OP** SpMV algorithm (see Algorithm 1.8). We thus adapted our simulation to use this algorithm for PHI. To ensure a fair comparison, we allocated the same amount of memory to PHI as for SpDC and the baselines (32 KiB). In Table 3.2, we report the results for PHI alongside our previous results (columns 2, 3, 6 and 7).

As expected, PHI achieves a significant reduction in memory traffic compared to the GC, down to 50.4% on average. This represents a 2.0× decrease compared to the GC, which is consistent with the results reported in the original PHI paper [41], and a 1.8× decrease compared to the RedC. However, SpDCache still outperforms PHI, cutting memory traffic by 2.1×. The improved performance of SpDC can again be explained by the better utilization of the cache-lines, with 7.21 useful words per evicted line, compared to 5.00 for PHI. The overall cache memory utilization is also significantly better with SpDC, with 75.2% of the cache holding useful data, compared to only 30.6% for PHI.

As explained in Chapter 2, PHI mixes the 'immediate updates' and 'deferred updates' strategies for **OP** SpMV. In the first phase of the algorithm, it processes *RED* transactions as 'immediate updates', and PHI handles them with a mechanism similar to a classical *reduction*

Table 3.2: All Results from Python Simulation

GMeans (48 mat.)	**GC**	**RedC**			**PHI**	**SpDC**					
sets GRC	*128*	*32*	*8*	*4*	*128*	*32*	*8*	*8*	*4*	*4*	*4*
sets DRC	*Ø*	*96*	*120*	*124*		*64*	*104*	*112*	*92*	*116*	*122*
sets SRT	*Ø*	*Ø*	*Ø*	*Ø*		*32*	*16*	*8*	*32*	*8*	*2*
Output Memory Traffic (% over GC)	100 ■1.0	89.2 ▼1.1	81.6 ▼1.2	80.3 ▼1.3	50.4 ▼2.0	24.5 ▼4.1	24.3 ▼4.1	25.5 ▼3.9	23.2 ▼4.3	25.3 ▼4.0	32.0 ▼3.1
Ratio of RMWs in Memory Traffic (%)	Ø	Ø	Ø	Ø	Ø	40.5	35.7	36.5	36.5	36.2	41.1
Total Memory Traffic (% over GC)	100 ■1.0	93.9 ▼1.1	91.9 ▼1.1	93.2 ▼1.1	98.8 ■1.0	56.1 ▼1.8	56.6 ▼1.8	56.9 ▼1.8	57.7 ▼1.7	58.2 ▼1.7	59.2 ▼1.7
Evicted Cache Line Utilization (max: 8)	3.60 ■1.0	3.56 ■1.0	3.66 ■1.0	3.67 ■1.0	5.00 ▲1.4	7.21 ▲2.0	7.21 ▲2.0	7.23 ▲2.0	7.22 ▲2.0	7.22 ▲2.0	7.25 ▲2.0
Cache Memory Utilization (%)	36.4 ■1.0	45.8 ▲1.3	47.0 ▲1.3	47.1 ▲1.3	30.6 ▼0.8	75.2 ▲2.1	68.3 ▲1.9	65.0 ▲1.8	72.5 ▲2.0	65.1 ▲1.8	60.9 ▲1.7

cache. The main differences are that the *GRC* and *DRC* are merged, which can cause unnecessary evictions of useful data, and more importantly, PHI uses a batching mechanism at eviction to 'defer' some updates to the second phase. When a cache-line is evicted and contains fewer non-zero entries than a fixed threshold, PHI batches the entries in *partial outputs* (*POs*) defined by the software and stored in memory. These *POs* are processed in the second phase of the algorithm, as in the 'deferred updates' variant. If the threshold is set too high, PHI batches many entries and increases second-phase memory traffic. If it is set too low, PHI batches few entries, reducing batching effectiveness and increasing the number of 'immediate updates', making PHI behavior closer to a simple *reduction cache*. PHI sets this threshold to one non-zero word per eight words in a cache-line, meaning that for each batched evicted line, PHI reduces bytes transferred by 4× (4 (word, address) tuples fitting in a single 64 B cache-line, instead of 4 full 64 B lines with 7 unused words each). However, we observed that, on the geometric mean over the 48 matrices, only 4.2% of PHI's output memory traffic comes from these batched updates (maximum 28.6%), indicating that PHI's overall behavior remains close to a simple *reduction cache*. In contrast, SpDCache addresses the drawbacks of a simple *reduction cache* by using dedicated regions for denser and sparser accesses, avoiding unnecessary evictions, without incurring the additional second-phase traffic introduced by PHI's batching mechanism.

We also explored different region sizing for SpDCache, varying the sizes of the *GRC*, *DRC* and *SRT* while keeping the total cache size constant at 32 KiB. The results are also reported in Table 3.2. Reducing the size of the *GRC* from 8 KiB (32 *sets*) to 2 KiB (8 *sets*) and 1 KiB (4 *sets*) has little impact on performance, as the *GRC* is mainly used for temporary storage of input vector elements. However, reducing it under 1 KiB would increase the total memory traffic, as more frequent evictions would occur before the data is used, an effect already noticeable when going from 2 KiB to 1 KiB. With constant *GRC* size, when reducing the size of the *SRT*, which means allocating more space to the *DRC*, we observe a slight increase in memory traffic, as the *SRT* becomes less effective at capturing irregular accesses. Thus, the *SRT* sizing appears to be more critical than the *DRC* sizing. We will explore the reasons behind this effect in the next chapter. Overall, fine tuning the region sizes can lead to small performance improvements, but the original sizing of 8 KiB for the *GRC*, 16 KiB for the *DRC* and 8 KiB for the *SRT* appears to be a good compromise for the set of matrices we evaluated.

We also explored different sizing strategies for the RedC baseline, varying the size of its single *reduction* region while keeping the total cache size constant at 32 KiB. Allocating more than 75% of the cache (96 *sets*) to store *RED* transactions slightly improves performance, as it increases the likelihood of capturing multiple updates to the same output element before eviction, without overly compromising the storage of input elements.

CONCLUSION

SpDCache outperforms PHI, a state-of-the-art *reduction* cache, by achieving a 2.1× decrease in memory traffic, thanks to its dedicated regions for in- and out-of-band accesses that improve cache-line and overall cache utilization. Additionally, we found that the simple 25/50/25% region sizing of SpDCache provides good performance with only minor variations observed when adjusting the sizes of the *GRC*, *DRC*, and *SRT*.

3.8 Conclusion

We have presented the concept of SpDCache, a *reduction cache* that has optimized storage and compute techniques for the sparser and denser regions of banded matrices. This approach does not reduce the number of compute operations, however, it significantly decreases memory traffic ($4.1\times$). These benefits can be attributed to three techniques: (1) blocking the dense part of the band in the *DRC*, (2) using fine-grained storage in the *SRT* and (3) off-loading the *reductions* for isolated elements to the downstream caches, to prevent local evictions and reduce traffic.

Exploiting the coarse-grain structure of matrices is key to design hardware optimized for sparse matrix kernels. In the next chapter, we demonstrate this with a more theoretical approach, allowing us to go further into SpDCache analysis, and justify appropriate region sizing, before going into a cycle-accurate evaluation.

TAKEAWAYS

Reduction cache:

- Support a *reduction* instruction: $RED(key, value)$
- Accumulate near memory
- Reduce the number of memory transactions

SpDCache:

- Take advantage of coarse grain structure of matrices: matrix regions such as *IBR* and *OBR* for banded matrices
- Three cache memory regions:
  - *GRC*, generic region cache for generic instructions
  - *DRC*, dense region cache for *reductions* in the *IBR*
  - *SRT*, sparse region table for *reductions* in the *OBR*
- Adapted memory transactions: cache-line wise versus element wise for the *DRC* and the *SRT*, respectively
- Overall: greatly reduce memory traffic

Chapter 4

Analysis of Reduction Cache Behavior and Parameter Exploration

Contents

4.1 Introduction

SPARSE matrix processing exposes hardware to irregular memory access patterns, which degrade cache efficiency and increase memory traffic. The magnitude of these effects on memory system performance remains challenging to predict due to the diversity of data patterns. In this chapter, we develop a theoretical model to analyze and optimize the behavior of a *reduction cache* when executing an *Outer-Product* (**OP**) SpMV kernel.

We propose a hybrid probabilistic/fast simulation model that predicts the memory transactions of a *reduction cache* during **OP** SpMV ($A \times b = c$). This model allows us to quickly explore the impact of different cache configurations on memory traffic, using banded matrices as a representative case study. By isolating the sequential reads of the input data and analyzing the density distribution of non-zero elements, we identify key design principles for optimizing cache performance.

The chapter demonstrates the importance of separating the in-band and out-of-band regions of the matrix into dedicated cache regions. This separation allows the cache to exploit the structural regularity of banded matrices while efficiently handling irregular accesses in sparser regions. The theoretical analysis validates the architectural choices of SpDCache and provides insights for sizing and configuring cache regions to minimize memory traffic.

In our analysis of *reduction caches*, we use a set-associative configuration, with 8 byte elements which we consider as the basic memory unit. s_C is the size of the cache, in elements. The cache is divided into *sets*, *ways* and cache-lines, of size S, W and l, respectively. Thus, $s_C = S \times W \times l$ elements. We recall that a banded square matrix of dimension M is define by its band width w_B, or its half-band width w_{hB}, such that $w_B = 2w_{hB} - 1$. The in- and out-of-band densities are named d_{IB} and d_{OB}, respectively. When d_{OB} is down to 0, the matrix is identified as 'perfectly' banded.

4.2 Isolating the Input Reads in a Generic Cache

WHEN computing an **OP** SpMV, all the irregular accesses come from updating the output vector c. However, the input matrix A and vector b are read sequentially, thus being efficiently cached by a generic cache that naturally exploits spatial locality.

With the matrix A stored in CSC format, each iteration of the **OP** algorithm necessitates reading four tables: $A.ptr$, $A.idx$, $A.val$ and b. In the same iteration, a fifth table is updated "randomly": c. If we consider cache-lines of size l elements, then each cache-line read of the $A.ptr$ table brings l consecutive entries, which are pointers to l consecutive columns of the matrix. Similarly, each cache-line read of the b table brings l consecutive entries, which will be multiplied by the non-zero elements of l consecutive columns of the matrix. These reads are needed in the outer-loop of the **OP** algorithm. In the inner-loop, each cache-line read of the $A.idx$ and $A.val$ tables brings l consecutive non-zero elements of the matrix to process, which will generate l "random" updates to the output vector c, thus l entire cache-lines of c may be accessed in the worst case.

To avoid unnecessary evictions, at each iteration, the generic cache should obviously keep $A.ptr$, $A.idx$, $A.val$ and b cache-lines up until they are no longer needed. That means l iterations for $A.idx$ and $A.val$, and around $l \times nnz_c$ iterations for $A.ptr$ and b. However, the cache must also keep the cache-lines of c that are being updated, which, over the course of $l \times nnz_c$ iterations, can necessitate up to the same amount of different cache-lines in the worst case, on unpredictable addresses. Thus, it would be difficult for a generic cache to keep all the useful data until it is no longer needed. Especially $A.ptr$ and b cache-lines, which are accessed once every nnz_c iterations, making eviction policies such as *LRU* unaware of their future use.

By isolating the input reads in a dedicated cache, as proposed with SpDCache and the *reduction cache* baseline in the previous chapter, we greatly increase the chances that all the cache-lines of A and b remain in the cache until they are no longer needed.

With a dedicated cache region for the inputs, the *GRC* in previous chapter, only the four sequentially read tables need to be considered. Each of these tables need to be read only once. With a classical *LRU* policy, $A.ptr$ and b cache-lines have lower chances of being evicted since the only accesses occurring while processing a column are the $\sim \frac{nnz_c}{l}$ cache-line reads of $A.idx$ and $A.val$. $A.idx$ and $A.val$ themselves can not be undesirably evicted anymore.

To avoid evicting $A.ptr$ and b too soon, the *GRC* must have enough size to handle the $\frac{nnz_c}{l}$ cache-line reads while computing a column. Thus, we define in Equation 4.1 the minimum size of the *GRC* in elements, which would be needed to achieve best performance in memory traffic.

$$s_{C_{min}}(GRC) = 2 \times l + 2 \times nnz_c \tag{4.1}$$

This relationship is not exact, as it does not consider address alignment effects, but it gives a good approximation of the minimum size needed for the *GRC* to avoid unnecessary evictions. It explains why, in the previous chapter, the total memory traffic slightly drops when the *GRC* size is decreased.

Separating the inputs in a dedicated cache region also has the advantage of allowing prefetching techniques [47] to be used efficiently. It has no impact on memory traffic but can improve performance by hiding memory latencies.

CONCLUSION

To avoid unnecessary evictions, it is profitable to **separate** the sequential reads of the inputs A and b from the "random" updates of the output c, into **dedicated cache regions**. A generic cache of reasonable size can achieve **low memory traffic** when processing A and b.

4.3 Model for Reduction Cache

To go beyond simply observing the behavior of a specific accelerator processing a given matrix, a model of the matrix and the cache is required. We present a hybrid probabilistic/fast simulation model for a *reduction cache*, processing **OP** SpMV, which predicts the volume of memory traffic issued by the L1 cache, starting from a matrix density distribution and a cache configuration.

The model represents the non-zero density both in the matrix and in the cache, with cache-line granularity. Using these coarse grain densities, a fast simulation of the **OP** SpMV is performed. The state of the cache can be predicted at each step of the simulation. By counting the evictions, we evaluate the output memory traffic, the memory trafic of the output vector c. We can thus predict the total number of memory accesses for different types of matrices and quickly explore different *reduction cache* configurations.

We do not consider the sequential accesses to the input matrix A and vector b in our model. Indeed, the matrix tables val, idx and ptr, and the vector b are each read once, giving a total memory traffic of $2nnz + (M + 1) + M$ elements. We showed in previous section that a simple generic cache of reasonable size is enough to handle the inputs.

Our model uses the cache configuration parameters S, W and l. Although our model is not limited to banded matrices, they are the focus of this work. The matrix density distribution thus takes on two different values, either d_{IB} in the band, or d_{OB} outside of the band.

We normalize the memory traffic by dividing the number of elements that are read or written by the number of non-zero elements (nnz) in the matrix, so we can compare performance between matrices of different sizes and densities. Indeed, we can interpret the normalized memory traffic, $normMT$, as the number of elements transferred from/to memory per non-zero matrix element. If the memory traffic remains constant while the matrix density increases, we expect $normMT$ to decrease.

4.3.1 Operation of the Analytical Model

Matrix Representation The square matrix is of dimension M, as represented in Figure 4.1. Its rows and columns are indexed by $0 \leqslant i, j \leqslant M - 1 \in \mathbb{N}$. Each column is split into vertical-strips of size l, where l is the cache-line size. Indeed a vertical-strip is a portion of a column j of size l, from row-indices i to $i + l - 1$, as illustrated by the *red* rectangle within the matrix in the figure. Within a column, vertical-strips are indexed by $k \in [\![0, M_l - 1]\!]$ [1], with $M_l = \lceil \frac{M}{l} \rceil$, and $k = \lceil \frac{i}{l} \rceil$.

The matrix is thus represented as a static, dense two dimensional table, MDT (*Matrix Density Table*), of size (M_l, M) that contains the density in $[0, 1]$ of each vertical-strip. We assume that the distribution of the non-zero elements within a vertical-strip is uniform, as

[1] We use $[\![a, b]\!]$ for the integer intervals, and $[a, b]$ for real intervals.

assumed in related work [51] where the authors studied a directly-mapped generic cache for an *Inner-Product* SpMV kernel, only for the case of 'perfectly' banded matrices.

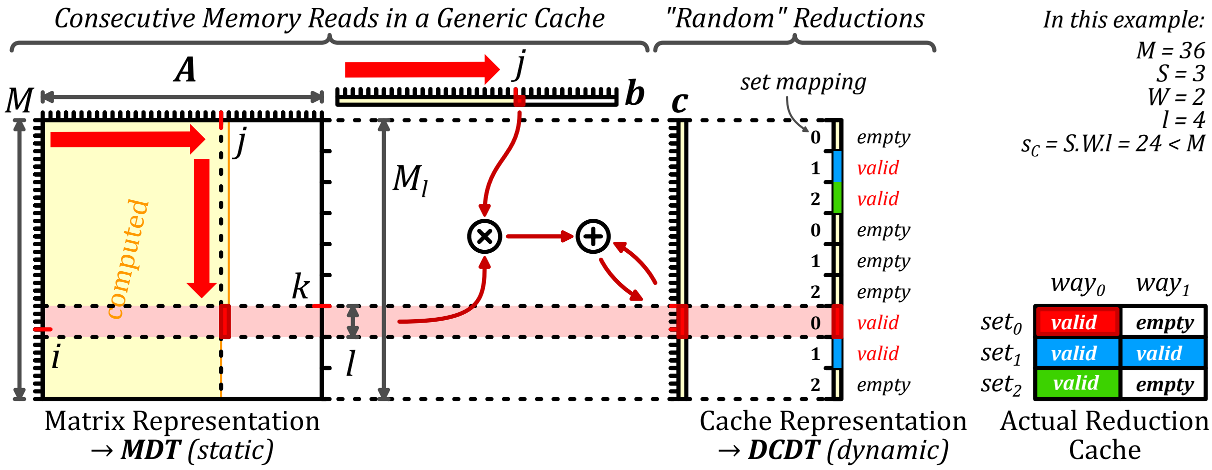


Figure 4.1: Input Matrix and Cache Representation of the Model

Cache Representation Still in Figure 4.1, the cache is represented as a single column, which maps directly to the output vector c. This "column-representation" of the cache is partitioned into cache-lines, corresponding directly to the vertical-strips of a given column of the matrix, as shown by the horizontal *light red* rectangle. A cache-line $k \in [\![0, M_l - 1]\!]$ is stored in a unique *set* $s \in [\![0, S - 1]\!]$ of the cache such that $s = k\%S$. Unlike the matrix representation which is static, the cache representation is continuously updated as the model evaluates **OP** SpMV.

Of course, the real cache-size ($S \times W \times l$) is not necessarily equal to the output vector size (M). Thus, at a given time, the number of valid (non-empty) cache-lines contained in the cache representation must not exceed the actual cache-size. Specifically, we ensure the number of valid cache-lines in a *set* of the cache representation does not exceed the number of ways W. In the example in Figure 4.1, the actual cache is holding four valid cache-lines, which are the only valid cache-lines present in the cache representation.

To count the number of valid cache-lines, we further distinguish them from empty ones. For this reason, we set the cache-line densities to be either zero (the cache-line is empty), or the density is in the range $\left[\frac{1}{l}, 1\right]$ (the cache-line is valid). A valid cache-line has a density between $\frac{1}{l}$ and 1, corresponding to the probabilistic number of non-zero elements.

The state of the cache is thus represented with the dynamic table $DCDT$ (*Disjoint Cache Density Table*), of dimension M_l that contains the disjoint-density in $\{0\} \cup \left[\frac{1}{l}, 1\right]$ of each cache-line, at a given iteration. Initially, all cache-lines are set to zero density, since the cache is considered empty.

Iterative Process In the fast simulation phase, the analytical model [18] iterates over the vertical-strips in the matrix, following the dense traversal of the **OP** algorithm, as described in Chapter 1 and shown with *red* arrows in Figure 4.1. In a given iteration (column j, vertical-strip κ), we use the matrix density of the current vertical-strip, $MDT_{\kappa,j}$, and the current state of $DCDT$ to determine if an eviction occurs, and update it for the next iteration, if necessary.

Disjoint-Density of the Current Vertical-Strip The density of the current vertical-strip in the matrix has to be disjointed at each iteration, to determine whether the strip is empty or not, as done for the cache-lines. We define the vertical-strip disjoint-density as $d_{lM}(\kappa) \in \{0\} \cup [\frac{1}{l}, 1]$.

- When $MDT_{\kappa,j} = 0$, the vertical-strip in the matrix is empty, thus, $d_{lM}(\kappa) = 0$.
- When $0 < MDT_{\kappa,j} < \frac{1}{l}$, the vertical-strip can have either 0 or 1 non-zero elements. To resolve this, we maintain a running counter of the accumulated $MDT_{\kappa,j}$ values, and when this counter exceeds $\frac{1}{l}$, the current vertical-strip is considered non-empty, so we set $d_{lM}(\kappa) = \frac{1}{l}$. If the counter is below this threshold, $d_{lM}(\kappa) = 0$.
- When $MDT_{\kappa,j} \geqslant \frac{1}{l}$, the vertical-strip has one or several non-zero elements. Thus, $d_{lM}(\kappa) = MDT_{\kappa,j}$.

With this definition, we can distinguish empty ($d_{lM}(\kappa) = 0$) and non-empty ($d_{lM}(\kappa) \neq 0$) vertical-strips.

Eviction Processing With the above definitions, we can determine if a miss occurs at the current iteration. A miss occurs when the cache-line κ is not valid and the vertical-strip is not empty. Recall that in a *reduction cache*, a miss without eviction results in the allocation of a free cache-line, initially filled with zeros, and produces no memory traffic. We define $miss(\kappa) = (DCDT_\kappa = 0)$ and $(d_{lM}(\kappa) \neq 0)$, a boolean indicating whether a miss occurs in the current iteration.

When allocating a free cache-line, it is necessary to ensure that the number of valid cache-lines in the *set* $s = \kappa\%S \in [\![0, S-1]\!]$ does not exceed W. The new allocation may thus trigger an eviction. Stated formally, $nnzcl_\kappa$ is the count of the cache-lines k from the *set* s that verify $DCDT_k \neq 0$. We define $evict(\kappa) = miss_\kappa$ and $(nnzcl_\kappa = W)$, a boolean indicating whether an eviction occurs in the current iteration.

In the case of an eviction, for the probabilistic model, we use a *random* eviction policy to select a valid cache-line in the *set* s (when $DCDT_k \neq 0$). The index of the evicted cache-line is $e \in [\![0, M_l - 1]\!]$.

Cache Update The last step is to update $DCDT$ before proceeding to the next iteration. (1) If a cache-line is evicted, we must account for the memory traffic and reset the victim line e; (2) The density of the current cache-line κ may increase if there is a hit or if it is newly allocated:

1. In the case of an eviction ($evict(\kappa)$ is true), we reset the victim cache-line (e) by setting $DCDT_e = 0$.
2. We update the density of the cache-line κ. In the case of a miss ($DCDT_\kappa = 0$), the cache-line density is set to the disjoint-density of the matrix vertical-strip κ. In the case of a hit, the cache-line and the vertical-strip densities are merged:
   $DCDT_\kappa \leftarrow DCDT_\kappa + (1 - DCDT_\kappa) \times d_{lM}(\kappa)$

The updated cache-line density is either 0 or greater than $\frac{1}{l}$, thus respecting the constraint that the cache-line is either empty or valid.

Final Iteration Starting with an initially empty cache, we run all $M \times M_l$ iterations and count the total number of evictions ($nbEvict$). After the last iteration, we account for the memory transactions required to flush the cache ($nbFlush$), writing the valid cache-lines to memory, by counting lines where $DCDT_k \neq 0$, for $k \in [\![0, M_l - 1]\!]$. $nbFlush$ will never exceed $S \times W$.

In a *reduction cache*, evictions and flushes result in a line-wise memory read, a modification, and a line-wise memory write. The update memory traffic is $2l(nbEvict + nbFlush)$ elements, and $normMT$ is this amount divided by nnz.

CONCLUSION

We have presented a **hybrid probabilistic/fast simulation analytical model** for a *reduction cache*, processing **OP** SpMV, which predicts the volume of memory traffic issued by the L1 cache, starting from a **matrix density distribution** and a **cache configuration**. The model represents the non-zero density both in the matrix and in the cache, with **cache-line granularity**. Using these coarse grain densities, a **fast simulation** of the **OP** SpMV is performed. In this way, we can quickly explore different *reduction cache* configurations to evaluate their impact on memory traffic.

4.4 Analysis of Reduction Cache Behavior

WE now study the *reduction cache* behavior when computing **OP** SpMV, starting with the simple case of 'perfectly' banded matrices with varying band widths. We then use the analytical model to analyse the *reduction cache* behavior, varying the out-of-band density and the cache-size. We show the benefit of storing the in- and out-of-band of the matrix in separate regions of the cache, to achieve low memory traffic.

4.4.1 Behavior of a Perfect-Band in a Cache

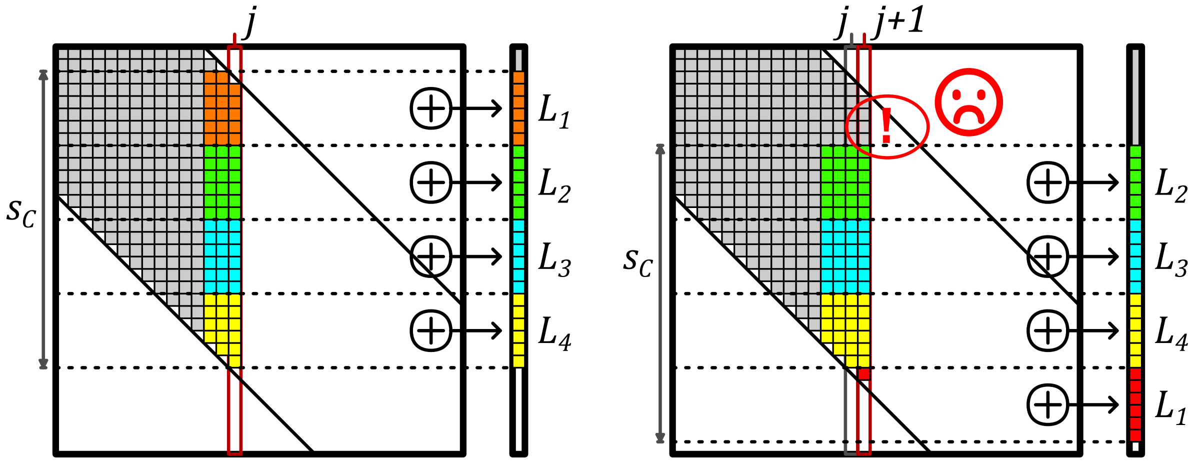
In the previous chapter, we presented SpDCache and a *reduction cache* baseline with a specific region (*DRC*) dedicated to storing the banded region of the matrix. We precised that the band of the matrix is chosen depending on the *DRC* size. In this section, we explain and precise this choice by analyzing the behavior of a *reduction cache* when processing an **OP** SpMV kernel with a 'perfectly' banded matrix.

We show that the *reduction cache* must be 'properly' dimensioned with respect to the band width of a 'perfectly' banded matrix, in order to avoid unnecessary evictions. Indeed, if the cache is too small compared to the band, evictions occur before the column iteration is over, leading to poor data-reuse for the next iteration. To achieve high data-reuse, we show that the cache-size (s_C) and the band width (w_B) must respect the relationship given in Equation 4.2.

$$s_C \geqslant w_B + l - 1 \tag{4.2}$$

We consider two examples with a cache having four cache-lines (L_1 to L_4) with $l = 6$. In Figure 4.2, we illustrate the case when the cache is too small ($s_C < w_B + 5$). In Figure 4.2a, we see the state of the cache after iterating up to column j. Then, in Figure 4.2b, we see the state after processing the next column, $j + 1$. The cache-line L_1 (*orange*) is evicted, and allocated to new elements at the bottom of the band, below the *yellow* area. In this case, there are still elements of the band above the *green* area, whose corresponding cache-line has, unfortunately, been evicted. Thus, on iteration $j+2$, cache-line L_1 must be fetched again and this pattern continues creating *thrashing*.

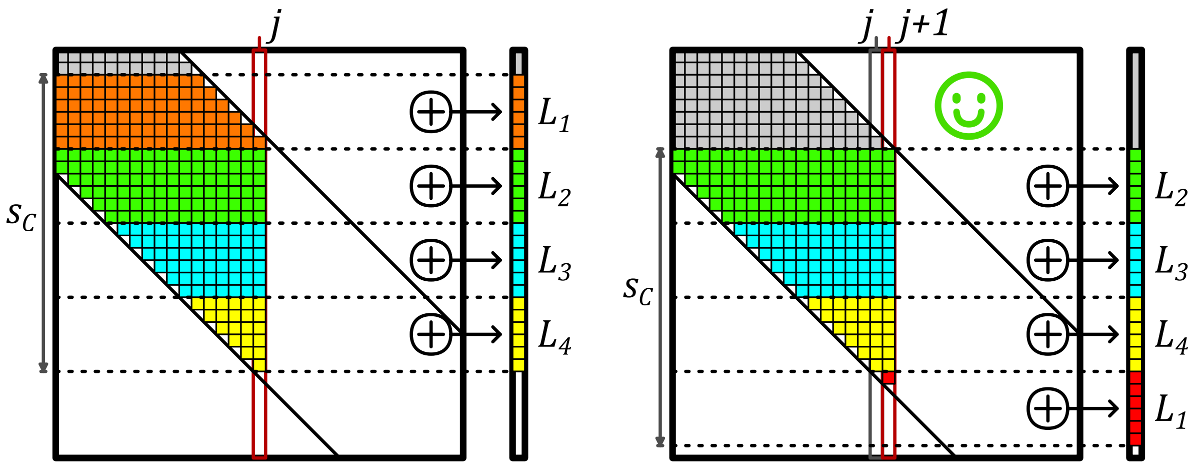
In Figure 4.3, we illustrate the case when the cache is 'properly' sized ($s_C \geqslant w_B + 5$). As in the previous example, Figures 4.3a and 4.3b show the state of the cache at iterations j and $j + 1$, respectively. This time, the cache-line L_1 is evicted only when all elements of the band



(a) Column iteration j **(b)** Column iteration $j+1$

Figure 4.2: Example of the Eviction Rotation in a Perfect Band when the Cache is Undersized ($s_C < w_B + l - 1$)

in the *orange* region have been processed. Indeed, the eviction of cache-line L_1 occurs just in time so that it can be allocated to the new region at the bottom of the band, below the *yellow* area. In this way, when processing columns, the cache-lines do a rotation where the top one is evicted and allocated at the bottom, with one eviction occurring after the processing of every l columns.



(a) Column iteration j **(b)** Column iteration $j+1$

Figure 4.3: Example of the Eviction Rotation in a Perfect Band when the Cache is 'Properly' Sized ($s_C = w_B + l - 1$)

If the cache-size is larger than required by Equation 4.2, the rotations are the same, the eviction-rate is the same, but each cache-line accumulates more elements, increasing the hit-rate. To produce this perfect cache rotation, we assume an LRU eviction policy and dense vertical-strips, although, even with a *random* policy, the general pattern still holds.

This analysis shows that, with a "properly-sized" *reduction cache*, a 'perfectly' banded matrix provokes no unnecessary evictions and the cache-lines are fully utilized. With a given cache-size, Equation 4.2 defines the maximum band width that can be efficiently stored in the *reduction cache* or in the *DRC* of SpDCache. For example, with a cache-size of $s_C = 2048$ elements (16 KiB) and a cache-line size of $l = 8$ elements, the maximum band width is $w_B = 2041$ elements (which means a half-band of $w_{hB} = 1020$ elements).

'Perfectly' banded matrices are rare in practice, especially if constraints by the above equation. In the next sections, we extend this analysis to banded matrices, studying the impact of out-of-band non-zero elements on the *reduction cache* behavior.

CONCLUSION

A *reduction cache* can achieve an efficient **rotating pattern** when processing a 'perfectly' banded matrix, provided that the cache-size and band width respect the **relationship given in Equation 4.2**: $s_C \geqslant w_B + l - 1$. This relationship defines the **maximum band width** that can be efficiently stored in the *reduction cache* for a given cache-size.

4.4.2 Impact of Out-Of-Band Density on Memory Traffic

We now apply the analytical model of Section 4.3.1 to analyse the performance of a *reduction cache* for banded matrices, processing an **OP** SpMV kernel ($A \times b = c$). We consider banded matrices of dimension $M = 1000$, with a half-band of size $w_{hB} = 125$ and an in-band density $d_{IB} = 1$. We analyse a 4-way set-associative cache ($W = 4$) with a cache-line size of $l = 8$ elements, with a *random* eviction policy. The size of the cache varies from $S = 1$ to $S = 32$ *sets*. When the cache-size reaches $S = 32$, the cache can hold the full output vector: $S \times W \times l \geqslant M$. In Figure 4.4, we plot the normalized memory traffic, $normMT$, for the output vector c, depending on the out-of-band density d_{OB}, for varying cache-sizes.

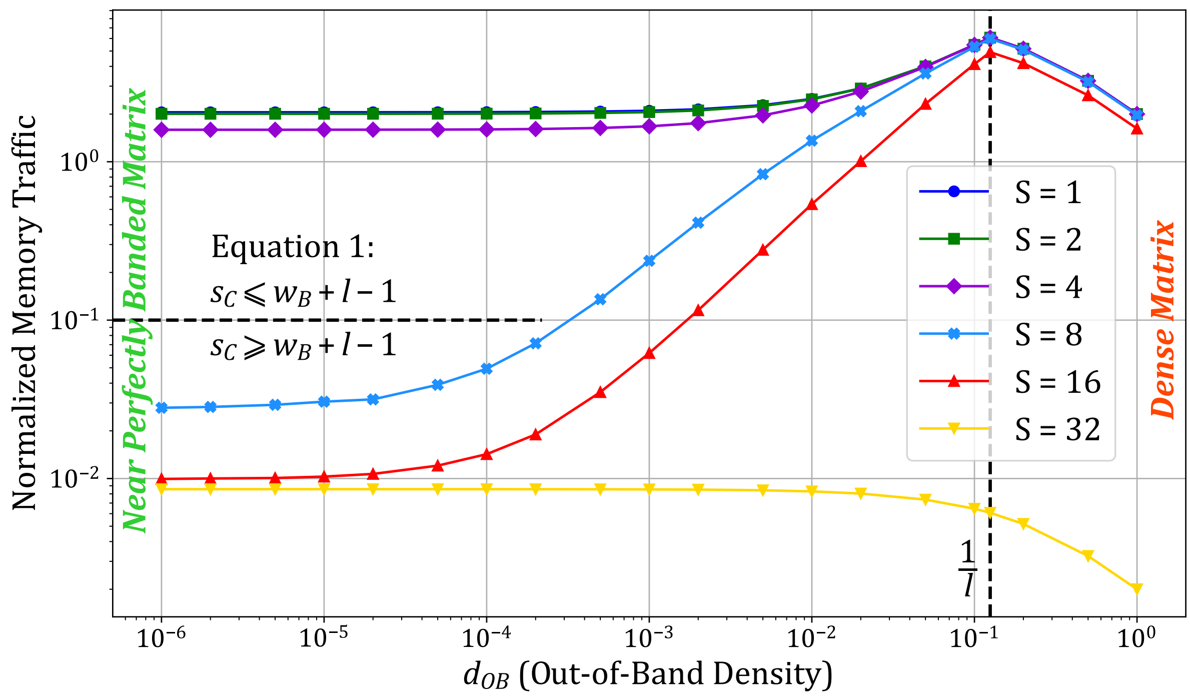


Figure 4.4: Normalized Output Memory Traffic Depending on the Out-of-Band Density for Several Cache-Sizes

On the right of the graph, we notice that $normMT$ reaches a peak when the out-of-band density equals $\frac{1}{l} = 0.125$. Indeed, when there is at least one non-zero element per cache-line, the behavior of the cache approaches that seen for a dense matrix. As the density increases further, the traffic remains constant, thus $normMT$ drops, as seen in the graph in Figure 4.4.

For the plot corresponding to $S = 32$, the cache is large enough to store the entire output vector c, thus, there are no evictions and $normMT$ is constant, corresponding to the flush traffic at the end. This plot gives the lower bound of memory traffic for the output vector c.

The first takeaway of this graph is that variations in d_{OB} heavily impact $normMT$. Isolated non-zero elements outside the band trigger misses and evictions of cache-lines corresponding to the banded region. Furthermore, when a cache-line is allocated for an isolated out-of-band element, most of the cache-line remains empty. In this case, instead of the efficient rotating pattern described in Section 4.4.1, the cache is *thrashing* with low utilisation. Regardless of the cache-size, the lower the out-of-band density, the lower $normMT$, as seen by the plots that tend downward toward the left of the graph. Avoiding the perturbations induced by the out-of-band elements would enable the cache to achieve minimal memory traffic, as with a 'perfectly' banded matrix.

Not surprisingly, at the left of the graph, $normMT$ varies depending on the cache-size. This is due to the relative size of the band and the cache. Indeed, if the matrix band width is large compared to the cache-size, processing one column of the band will generate many evictions. To avoid evictions in the band, the relationship given by Equation 4.2 in Section 4.4.1 must be fulfilled. Indeed, for this example, for $S \geqslant 8$, this relationship holds and the corresponding plots have significantly lower $normMT$, down by two orders of magnitude. Conversely, the first three plots ($S = 1$ to 4) do not respect this condition, explaining their high $normMT$. Thus, the second takeaway from this graph is that, given a cache memory size, there is an ideal band width that generates minimal $normMT$. Increasing the cache-size beyond this threshold only produces a minor reduction in memory traffic between $\sim 9 \times 10^{-3}$ and $\sim 2 \times 10^{-2}$, since at the beginning of matrix traversal, the first eviction is deferred before reaching the steady state rotating pattern, as discuss in Section 4.4.1.

These observations explains the need of dividing the *reduction cache* into two regions. One which is optimized for storing the band and the other which handles all the other elements. The in-band region of the cache can thus operate with maximal efficiency corresponding to the bottom-left of Figure 4.4. In subsequent sections, we analyse the out-of-band traffic.

CONCLUSION

The out-of-band density heavily impacts the normalized memory traffic $normMT$ as the isolated elements trigger unnecessary evictions, preventing the efficient rotating pattern shown in previous section. Thus, to minimize memory traffic, it is important to **separate** the band and out-of-band regions of the matrix into **dedicated regions** of the *reduction cache*.

4.4.3 Design Analysis for Out-of-Band Region

For the type of large matrices under study, there are usually too many non-zero elements in the out-of-band region for them to be cached from one column to the next, thus making it impossible to achieve the rotating pattern described in Section 4.4.1. Therefore, the size of the cache for the out-of-band is not as important as the choice of associativity and data granularity. To justify this affirmation, we use the analytical model from Section 4.3.1 to analyse the performance of the *reduction cache* when processing only the non-zero elements outside of the band. As we have already discussed the handling of the in-band region, we now consider only the out-of-band elements, which is equivalent to setting the in-band density to $d_{IB} = 0$. Using the same matrix parameters as in the previous section, we explore different choices of cache-line size ($l \in \{1, 8\}$ elements), associativity (set-associative or fully-associative), memory transaction type (*R/W* or *RMW*), and total cache-size ($s_C \in \{128, 256\}$ elements). Because the out-of-band data is highly sparse, we want to explore having an element-wise cache ($l = 1$) instead of the typical line-wise cache ($l = 8$). As this cache region can be relatively small, a fully-associative configuration ($S = 1$, $l = 1$) is reasonable in terms of hardware implementation, and this full associativity makes full use of the storage. Indeed, when dealing with sparse elements, it is wasteful to bring a full line into the cache to only modify one element and then evict it, which we denote as *R/W*. Thus, we consider offloading the *reduction* to a higher level cache, thus reducing unnecessary data transfers. We assume the higher level cache can support this operation which we denote as *RMW*, following the design choice of SpDCache in Chapter 3. We recall that protocols such as AXI [1] support atomic transactions (AMOs), which could easily be extended for *reduction* operations.

To analyse the impact of the cache-size, we consider two different sizes (*solid* and *dashed*). For a given cache-size, we adjust the number of ways (W), based on the associativity, to keep the cache-size constant. In Figure 4.5, we plot $normMT$ for the output vector c, depending on d_{OB}, for varying cache configurations.

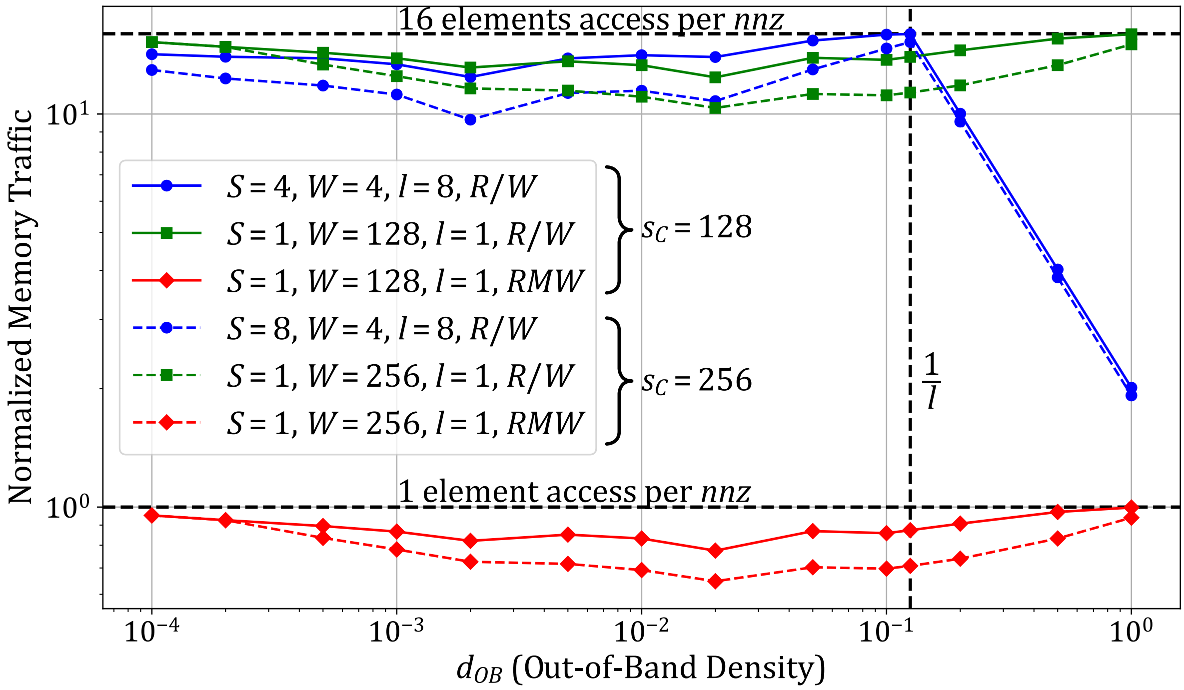


Figure 4.5: Normalized Output Memory Traffic of the Out-of-Band, Depending on the Density for Several Cache-Sizes

Let us start by considering a classic set-associative cache with a line-wise memory interface (*solid blue*, *dot*), where we see a dependence of $normMT$ on d_{OB}. On the right of the graph ($d_{OB} \geqslant \frac{1}{l}$), we observe the same behavior as in Section 4.4.2, where the cache behavior approaches that seen for a dense matrix, where the cache tends to *thrash*.

Looking at the plot for the fully-associative cache with a line-wise memory interface (*solid green*, *square*), we see that the normalized traffic is not impacted by the density of the out-of-band, as expected. With the fully-associative cache, $normMT$ is constant near 16, as, on average, there are two line-wise transactions per non-zero element processed, one read to fill the line, one write to evict it.

The *red* plots (*diamond*) correspond to a fully-associative cache with an element-wise memory interface, which can offload element-wise *RMW* transactions. We see that this approach results in over an order magnitude reduction in $normMT$. First, we avoid transferring full cache-lines which are mostly empty, and we only send the *RMW* in one direction ("fire-and-forget"). Indeed, $normMT$ is now reduced to below one transaction per non-zero element processed. The fact that this value is below one shows that this cache region does hit and reduce memory traffic. Increasing the cache-size from 128 (*solid*) to 256 (*dashed*) elements reduces $normMT$ by at most 10^{-1}, the hit-rate being slightly increased as expected.

CONCLUSION

For the out-of-band region of the cache, the most important characteristics are: (1) a **fully-associative** element-wise data-storage to avoid virtually empty cache-lines, and (2) an **element-wise memory interface** to reduce memory traffic. With these two characteristics, the normalized memory traffic can be reduced to below one transaction per non-zero element processed.

4.4.4 Impact of In-Band Density

In the previous analysis, we considered a dense band ($d_{IB} = 1$) to show the importance of having a rotating eviction pattern to decrease memory traffic. However, matrices from real applications do not have fully dense bands. Indeed, from our set of 48 matrices used in the previous chapter, considering a band width of $w_B = 2041$ that fits into a *DRC* of 16 KiB ($s_C = 2048$ elements), the average in-band density is $d_{IB} = 2.11 \times 10^{-3}$, oscillating between 2.00×10^{-8} and 5.44×10^{-2}, against an average out-of-band density $d_{OB} = 1.28 \times 10^{-4}$, oscillating between 0.00 and 1.04×10^{-2}. Considering that a part of the matrices of our set are not considered as banded, this order of magnitude difference between *dIB* and d_{OB} in average on all 48 matrices shows that choosing the band as a region partitioning method has great chances to pick up the denser region of any matrix.

Since the in-band density is lower than 1 in practice, we use the model to analyse the impact on normalized memory traffic of d_{IB} variations. We use the same parameters as in the previous sections, a matrix of dimension $M = 1000$ and a cache with four *ways* ($W = 4$) and cache-lines of size $l = 8$. We do not consider the out-of-band ($d_{OB} = 0$) since it is processed in a separate region. We focus on the in-band region for several cache configurations, making sure that the band width for each *set* size follows the relationship in Equation 4.2. In Figure 4.6, we plot $normMT$ for the output vector c, depending on d_{IB}, for varying cache configurations.

We observe that $normMT$ increases as d_{IB} decreases, generally following a constant negative slope (*dashed*) on the right side of the graph for each cache size. Thus, across a broad range of in-band densities, the raw memory traffic remains relatively constant. As noted in Section 4.4.2, the $S = 32$ curve (*yellow*) indicates the lower bound of memory traffic since the entire matrix can fit within the cache. In this scenario, there are no eviction in the band until the density drops sufficiently low ($< 10^{-3}$), resulting in isolated non-zero elements. For other cache sizes, variations around the constant raw memory traffic are observed. If a ideal eviction policy were in place, allowing the rotating pattern to function as intended (refer to

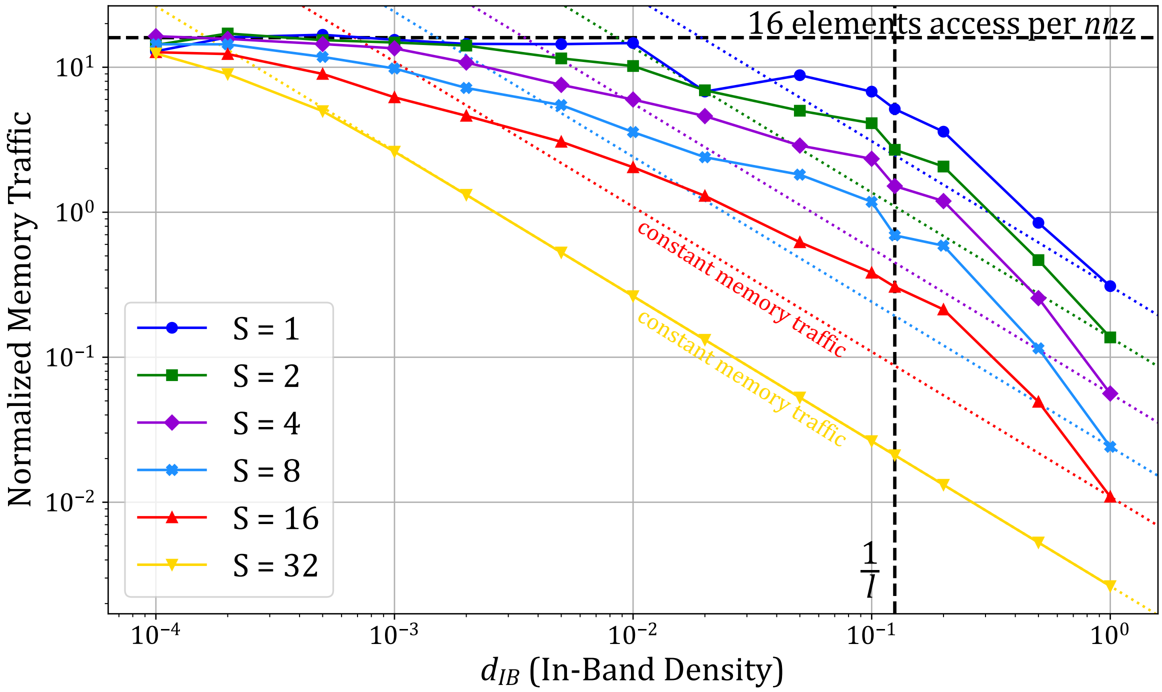


Figure 4.6: Normalized Output Memory Traffic of the In-Band, Depending on the Density for Several Cache-Sizes

Section 4.4.1), we would expect all curves to align with their respective *dashed* lines, indicating constant raw memory traffic. However, the model employs a *random* policy, which disrupts the rotating pattern and accounts for the observed deviations. An *LRU* policy would likely yield improved results, aligning with the constant line for densities above $1/l$ and deviating for lower densities. The optimal policy would prioritize evicting lower addresses (the top of the band), thereby maintaining the rotating pattern.

For low d_{IB} (left side of the graph), $normMT$ plateaus at 16 elements transferred per non-zero element for all cache sizes. With very low density the non-zeros are isolated and provide no reuse: each non-zero triggers a full cache-line read and a full-line write-back. With $l = 8$ this corresponds to 16 elements transferred per non-zero.

It is therefore important to select the densest region of the matrix to be processed in a dedicated cache region such as the *DRC* of SpDCache. However, the selected region need not be highly dense: for a 'perfectly' banded matrix with $M = 1000$, columns become nearly empty for $d_{IB} \lesssim 10^{-3}$, yet the raw memory traffic remains nearly constant in the range 10^{-2} to 10^{-3} depending on cache size. Because real, larger matrices often have a mean d_{IB} around 10^{-3}, a *DRC* sized to capture this denser region still provides good memory-traffic performance.

NOTE

In the SpDCache architecture, the *DRC* employs an *LRU* or pseudo-*LRU* eviction policy instead of an ideal custom policy. Despite this, we show that these simple, well-known policies still deliver good performance.

CONCLUSION

The **in-band density** has only a **marginal impact on memory traffic**, with real matrices typically exhibiting in-band densities around 10^{-3}.

4.4.5 Resulting Approach and Discussion

In the previous sections, we analyzed the behavior of *reduction caches* executing an **OP** SpMV kernel with banded matrices. We evaluated the expected memory traffic using an analytical model, considering separately the in- and out-of-band regions. We identified the cache characteristics required for both regions, which are quite different. We thus validate the proposition to divide the available cache memory into two dedicated regions: one for the in-band, and one for the out-of-band; the *DRC* and *SRT* of SpDCache, respectively.

To obtain the efficient rotating pattern for the in-band region, which is the key to massively reducing memory traffic, Equation 4.2 must be respected. In practice, this defines the size of the band that can be allocated to the in-band cache region. As we have seen, this cache region can operate efficiently with a set-associative line-wise configuration, on a wide range of in-band densities.

For the out-of-band region, a fully-associative element-wise cache with an element-wise memory interface can significantly reduce the memory traffic.

When sizing the two regions, once Equation 4.2 is satisfied, enlarging the in-band region yields only a marginal reduction in memory traffic (on the order of 10^{-2}). By contrast, a moderate increase of the out-of-band region produces a larger reduction (on the order of 10^{-1}). Therefore, the out-of-band region should be made as large as practical, subject to the hardware limits imposed by full associativity. This correlates with the results presented in Chapter 3, where we showed better performance with SpDCache configurations having a larger *SRT*.

Although this analysis is sufficient to guide the design of SpDCache for **OP** SpMV with banded matrices, it is important to note that the analytical model that we have presented is based on the assumption of a uniform distribution of the non-zero elements. In practice, this is not always the case, and this non-uniformity can impact the cache behavior, as will be seen later in Chapter 6. Indeed, if the non-zero elements are clumped, the cache-lines tend to be better filled, resulting in lower memory traffic. In other words, our uniform assumption leads to a pessimistic prediction of memory traffic.

CONCLUSION

To minimize memory traffic when processing banded matrices with a *reduction cache*, it is important to **divide** the cache into two dedicated regions: (1) an **in-band region** dedicated to a fixed band width of the matrix according to the Equation 4.2, and configured as a **set-associative line-wise** cache, and (2) an **out-of-band region** configured as a **fully-associative element-wise** cache with an **element-wise memory interface**. The **out-of-band region** should be made as large as practical, as it has a greater impact on memory traffic than enlarging the in-band region.

4.5 Conclusion

This chapter presented a theoretical model to analyze the behavior of *reduction caches* when processing sparse matrices, specifically focusing on the **OP** SpMV kernel. Through a hybrid probabilistic/fast simulation model, we demonstrated how the structure of banded matrices can be leveraged to optimize cache performance.

Our analysis highlighted the importance of separating the in-band and out-of-band regions of the matrix into dedicated cache regions. For the in-band region, a set-associative, line-wise cache configuration ensures efficient data reuse by maintaining a rotating eviction pattern, minimizing unnecessary memory transactions. For the out-of-band region, a fully-associative, element-wise cache with an element-wise memory interface significantly reduces memory traffic by avoiding the transfer of mostly empty cache-lines.

The theoretical insights gained from this model validate the architectural choices of SpDCache, particularly the use of distinct regions for denser and sparser data. These findings provide a robust foundation for designing and sizing cache regions to minimize memory traffic, ultimately improving the efficiency of sparse matrix computations. In the next chapter, we will delve into the microarchitectural details of SpDC, translating these theoretical insights into practical hardware design.

TAKEAWAYS

Simulating *reduction caches* and SpDCache

- **High level probabilistic/fast model in C** (4.3):
  - fast simulation, over generic matrix density distributions, easy parameter exploration, deep understanding of cache behaviors, but,
  - very high level (some approximations and hypothesis), limited on eviction policies.
- **Transaction level model in Python** (3.6):
  - faster than a full RTL simulation, accurate performance on memory traffic, functional testing, but,
  - slower than the probabilistic model, demands real matrices in input, two factors that alleviate the parameter exploration capabilities, no latency performance compared to RTL simulation.
- **RTL implementation** (next chapter):
  - accurate results in both memory traffic and latencies, hardware testing, but,
  - slow simulation, demands real matrices in input, time-consuming implementation.

Validation and sizing of SpDCache three regions

- ***GRC***: can take advantage of prefetching methods, size flexible.
- ***DRC***: must respect relationship $s_C \geqslant w_B + l - 1$ (Equation 4.2), size flexible.
- ***SRT***: must be element-wise, especially on memory transactions, high size is better but limited by hardware constraint.

The most basic and simpler to implement region sizing for SpDCache, **25%/50%/25%** for *GRC*, *DRC* and *SRT*, respectively, is close to optimal for memory traffic performance.

Chapter 5

Microarchitecture and Evaluation

Contents

5.1 Introduction

TODO

5.2 Microarchitecture

OBaMMA features two primary capabilities. First, as a *reduction cache*, it supports *reduction* instructions, enabling the accumulation of data near-memory. Second, it implements a region-based cache where regions have different data-granularity, levels of associativity and overall behavior.

5.2.1 HPDcache: The RTL base generic cache

SpDCache is implemented into an L1 data cache for a CVA6 RISC-V core, building upon an existing open-source high-performance set-associative data cache known as HPDcache [19]. In contrast to traditional data caches, HPDcache introduces new features, including buffering techniques for memory writes, set-associative storage to manage read miss requests, and dynamic configurability of each cache-line to operate in either write-back or write-through modes. It is designed for high performance, utilizing a three-stage pipeline to handle multiple instructions while minimizing stall potential through out-of-order cache execution.

An overview of SpDCache's microarchitecture is shown in Figure 5.1, where *black* and *blue* components are originally from HPDcache. The first stage (*st0*) decodes instructions and issues reads to the cache directory and data RAMs (*Dir* and *Data*). These reads require one cycle to respond. In the second stage (*st1*), the response indicates whether a hit or miss occurred and if an eviction is necessary. The instruction proceeds to the final stage (*st2*) only if a miss or eviction must be processed. At *st0*, the instruction is not yet consumed. Depending on the response and the availability of cache resources at *st1*, a *nop* (*no operation*) may be inserted in the pipeline, and the instruction may be placed in a *replay table*. The instruction is then replayed with the highest priority once its dependencies are cleared. If the instruction does not need to be replayed at *st1*, it is consumed and processed.

HPDcache provides extensive static and dynamic configurability, making it a flexible foundation for further extensions. To implement SpDCache as well as a *reduction cache* baseline, we modified HPDcache to be highly configurable so that a single RTL implementation can produce multiple designs. The RTL can be instantiated as a generic cache (the first baseline, GC), as a *reduction cache* (the second baseline, RedC), or as SpDC, the solution presented in the previous chapters.

5.2.2 RTL Implementation of SpDCache

We configure the original HPDcache as a 4-way set-associative cache with 128 *sets* and 64-byte cache-lines, resulting in a total cache size of 32 kiB. We extend the three-stage pipeline to handle the new RED instruction. Figure 5.1 provides an overview of OBaMMA's microarchitecture, highlighting new blocks in *red*, modified blocks in *blue*, and untouched components in *black*. Let us briefly describe the operation of the DRC under three scenarios: hits, misses with available cache-lines, and misses requiring eviction.

In the case of a hit, the value of the RED instruction is accumulated with its corresponding value in the cache. At *st1*, the cache value, read in *st0* from *Data*, is accumulated with the RED value using a dedicated accumulator (*Acc*). The result is written back to *Data* at *st2*.

In the case of a miss, if there is an available cache-line, the value of the RED instruction is simply stored in this free entry of the cache. A free cache-line is selected at *st1*, and at *st2*,

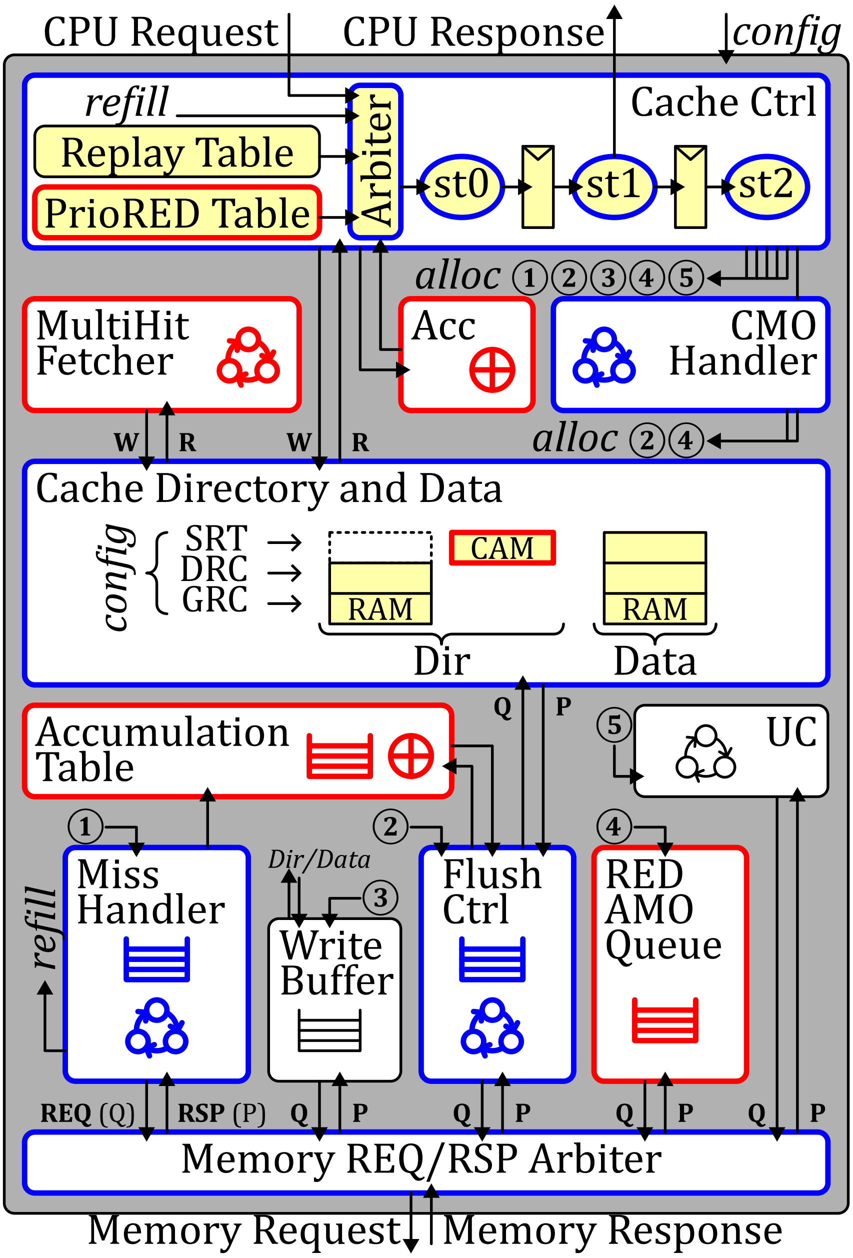


Figure 5.1: OBaMMA Microarchitecture, based on HPDcache

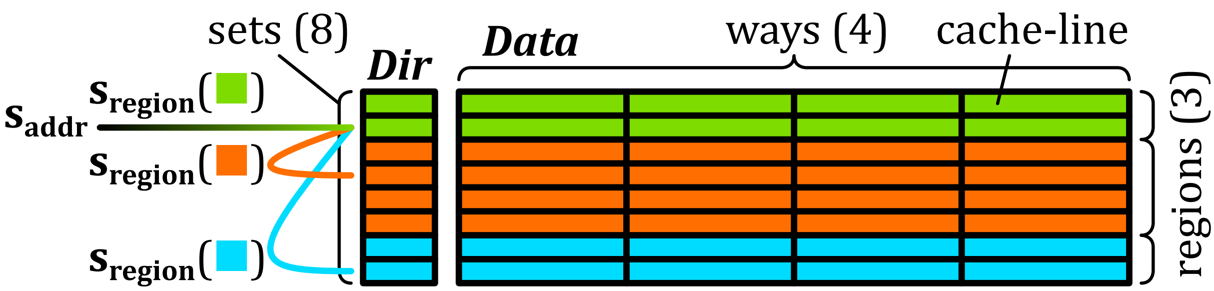


Figure 5.2: Set Redirection Depending on the Region

the RED value is written to *Data* along with zeros to reset the selected cache-line. A write to *Dir* is also performed to change the state of the cache-line from free to valid.

If a miss occurs but no cache-line is free, an eviction must be processed before handling the RED instruction. The RED instruction is placed in the *PrioRED table* to be replayed with higher priority than the *replay table*. Meanwhile, at *st2*, the *miss handler*, *flush controller*, and *accumulation table* are allocated to process the eviction. The eviction is a "fire-and-forget" operation, meaning the RED instruction only needs to wait for the entry in *Data* to be freed, not for the entire eviction to finish. The eviction process begins by fetching and resetting the chosen 64 B cache-line from *Data*, which is done in one cycle by the *flush controller*. Simultaneously, the *miss handler* issues a read to the main memory, which in our performance simulations, we assume will take 64 cycles to respond. The *accumulation table* waits and temporarily stores both the old cache-line from the *flush controller* and the 64 B response data from the *miss handler*. It then accumulates the cache-line with the data, word by word, and sends the 64 B results back to the main memory using the existing write interface.

With the changes described above, we have transformed HPDcache into a *reduction cache*. To achieve multiple cache regions, we virtually separate the cache RAMs. To maintain HPDcache's set-associative operation, we divide the cache memory along its *sets*. Thus, each region of the cache has the same structure as the overall cache but is smaller in size: each *set* contains a fixed number (the number of ways) of 64-byte cache-lines, as depicted in Figure 5.2. Each region can map any possible address. To ensure that every address matches a unique *set* within each region, we allocate a power-of-two number of consecutive *sets* to a region. Considering the total number of *sets* S_T and our region R mapping on $S_R < S_T$ of them, starting at the *set* s_{offset}, if an address matches one of the *sets* of this region, no action is required. However, if the address matches a *set* s_{addr} outside the region, we need to map it to a unique *set* in the region such that $s_{region} = (s_{addr}\%S_R) + s_{offset}$, as illustrated in Figure 5.2. The three regions GRC, DRC, and SRT are each defined by their number of *sets* S_R and their first *set*, s_{offset}. OBaMMA can thus be dynamically reconfigured to operate as a generic cache, a traditional *reduction cache* or a region-based *reduction cache* based on the needs of the application (*config* in Figure 5.1).

The SRT region requires additional adjustments to function as an element-wise fully-associative cache. This region operates as if all the *sets* were combined into a single *set*, containing numerous elements. Each element requires its own directory entry. Therefore, we replace the *Dir* RAMs of this region with a *CAM* (*Content-Addressable Memory*) containing the addresses of each valid element in the SRT region. With the *CAM*, we can check for a hit across the entire region and retrieve the matching value stored in the *Data* RAMs. We also added a queue (*RED AMO Queue*) to manage the element-wise read-modify-write memory transactions.

To avoid address duplication between regions, the *MultiHit fetcher* in Figure 5.1 manages coherency between the DRC and SRT. Since software selects the region, this can be necessary.

We have added several custom instructions to the CVA6 RISC-V core [**?**]. The primary addition is the *reduction* instruction (*RED*), which takes an address and a value as arguments and expects no response. This instruction is accompanied by a region instruction (*REG*) that specifies in which region subsequent *reductions* should be processed. We have also added an instruction to flush the *reduction* regions (DRC and SRT) of the cache, which is necessary at the end of the algorithm. It is performed in the cache by the *CMO handler*. Finally, we have included an instruction to enable and disable the *reduction* mode of OBaMMA.

5.3 Evaluation

TODO

Table 5.1: Results of Main Cache Configurations from Python Simulation

GMeans (48 mat.)		GC	RedC	SpDC
sets	GRC	*128*	*64*	*32*
	DRC	*Ø*	*64*	*64*
	SRT	*Ø*	*Ø*	*32*
Output Memory Traffic (% over GC)		100 ■1.0	103 ■1.0	24.5 ▼4.1
Ratio of RMWs in Memory Traffic (%)		Ø	Ø	40.5
Total Memory Traffic (% over GC)		100 ■1.0	98.5 ■1.0	56.1 ▼1.8
Evicted Cache Line Utilization (max: 8)		3.60 ■1.0	3.41 ■1.0	7.21 ▲2.0
Cache Memory Utilization (%)		36.4 ■1.0	43.8 ▲1.2	75.2 ▲2.1

5.3.1 Methodology

To evaluate our solution, we performed a full RTL implementation of OBaMMA. The starting point is a CVA6 [**?**] RISC-V core with HPDcache [19] as an L1 data-cache with size 32 KiB ($S = 128$, $W = 4$, $l = 8$ elements). This serves as our baseline for performance comparison which we refer to as the *Generic Cache* (*GC*). Next we evaluate a *reduction cache* which we derived from the HPDcache and configured to have two regions: a generic region with 16 KiB, and a *reduction* region with 16 KiB whose behavior is the same as the DRC region of OBaMMA. We refer to this design as the *Reduction Cache* (*RedC*). Finally, we evaluate OBaMMA with a reference configuration of: GRC = 8 KiB, DRC = 16 KiB, SRT = 8 KiB. The configurations are summarized in Table 5.2. All our simulations are performed using Siemens Questasim™ 2023.3 and the memory accesses beyond the L1 incur a fixed latency of 64 clock cycles.

Table 5.2: Region Sizing of the Cache Configurations

Configuration	GRC	DRC	SRT
GC	100% ($S = 128$)	0% ($S = 0$)	0% ($S = 0$)
RedC	50% ($S = 64$)	50% ($S = 64$)	0% ($S = 0$)
OBaMMA	25% ($S = 32$)	50% ($S = 64$)	25% ($S = 32$)

By default, matrices are stored in CSC sparse format. The code running on the CVA6 is written in generic C, compiled with *gcc* 11.2.0 and -O3 optimization. The inner-loops of the **OP** SpMV kernel are unrolled 4 or 8 times and the RED instructions are manually inserted using *asm* directives. For OBaMMA, it is necessary to know the target region for each RED instruction based on the size of the band. The value of w_{hB} is calculated using Equation **??**

Listing 1: C Code for the **OP** SpMV with RED Operations

```
int OBtop = PTR[0];
for (int n=0, int k=0; n < N; n++, k+=3) {
    xn = X[n];
    int pos     = OBtop;    /* start of column */
    int IB      = PTR[k+1]; /* start of band */
    int OBbot   = PTR[k+2]; /* end of band    */
    int nextCol = PTR[k+3]; /* end of column */
    /* 1st step: top out-of-band region */
    REG(SRT); /* set region = SRT */
    blockPos = (IB - OBtop) / 8;
    for (int i=0; i < blockPos; pos+=8, i++)
        /* Read IDX, VAL, Mult., RED unroll 8x */
        REDiter_unroll_8(pos, xn);
    for (; pos < IB; pos++)
        REDiter_1(pos, xn);
    /* 2nd step: in-band region */
    REG(DRC); /* set region = DRC */
    blockPos = (OBbot - pos) / 8;
    for (int i=0; i < blockPos; pos+=8, i++)
        REDiter_unroll_8(pos, xn);
    for (; pos < OBbot; pos++)
        REDiter_1(pos, xn);
    /* 3rd step: bottom out-of-band region */
    REG(SRT); /* set region = SRT */
    blockPos = (nextCol - pos) / 8;
    for (int i=0; i < blockPos; pos+=8, i++)
        REDiter_unroll_8(pos, xn);
    for (; pos < nextCol; pos++)
        REDiter_1(pos, xn);
    OBtop = nextCol;
}
flush();
```

so the in-band region fits in the DRC and is independent of the matrix. In our reference configuration ($s_C(DRC) = 2048$, $l = 8$), $w_{hB} = 1021$.

The region for the RED instructions could be calculated on-the-fly, however, this introduces additional instructions in the critical inner-loop. To avoid this overhead, the region is pre-computed and stored in memory using our BaCSC format (see Section **??**). At the start of each column, in the outer-loop, three pointers are looked-up to identify the the top out-of-band region, the in-band region and the bottom out-of-band region. Thus, we can process these three regions of the column in three separate, consecutive inner-loops, which do not require additional instructions to determine the region. The main loop of our modified **OP** SpMV is shown in Listing 1.

5.3.2 Matrix Selection

For our evaluation, we have selected an extended set of 62 square matrices from the SparseSuite Matrix Collection [**?**] which includes virtually all the matrices used by recent hardware accelerators [42, 48, 59, 61, 62]. We verified that this set is diverse containing banded and non-banded sparse matrices with various sizes ($M \in [124, 1\ 971\ 281]$), numbers of non-zero elements ($nnz \in [2\ 157, 14\ 148\ 858]$) and a wide range of densities ($d \in [1.4 \times 10^{-6}, 7.0 \times 10^{-1}]$).

Based on our experimental results, we observe that these matrices could be divided into two groups based on their memory performance characteristics. We define a new metric, d_{cd}, which is the average density of the cache-line sized clusters in the columns of the entire matrix. When this metric is low ($d_{cd} \ll \frac{1}{l}$) the DRC is less effective and the problem tends to be memory-bound. When this metric is high ($d_{cd} \gg \frac{1}{l}$) there is significant spatial reuse

Table 5.3: Characteristics of Selected Group 1, 2 Matrices

ID	Matrix	M	nnz	Density	nnz/C	d_{cd}
			Group 1			
11	poisson3Da	13.5k	352k	1.93e−3	26.1	14%
12	cit-HepTh	27.8k	353k	4.57e−4	12.7	13%
13	soc-Epinions1	75.8k	508k	8.84e−5	6.7	16%
14	patents_main	241k	561k	9.69e−6	2.3	12%
15	sme3Db	29.1k	2.08M	2.46e−3	71.5	15%
16	Stanford	282k	2.31M	2.91e−5	8.2	12%
			Group 2			
21	scircuit	171k	959k	3.28e−5	5.6	24%
22	msc10848	10.8k	1.23M	1.05e−2	113.4	73%
23	opt1	15.4k	1.93M	8.09e−3	125.0	86%
24	xenon2	15.7k	2.87M	1.56e−4	24.6	40%
25	consph	83.3k	6.01M	8.66e−4	72.1	57%
26	x104	108k	10.0M	8.66e−4	93.8	78%

in the cache and the problem tends to become compute-bound. We have set the threshold between these two groups at $\frac{1.5}{l} = 18.75\%$ as this corresponds to the point where, on average, there is more than one element per cache-line, which is sufficient for a scalar core. Based on this threshold, there are 20 matrices in group 1 ($d_{cd} \leqslant 18.75\%$) and 42 in group 2 ($d_{cd} > 18.75\%$). Within each group, we selected six well-known matrices which have often been used for performance evaluation as shown in Table 5.3. We note that the 12 selected matrices are representative of different overall matrix densities ($nnz/Column$) and d_{cd} values as shown by the *red* points in the scatter plot in Figure 5.3. We report: (1) the performance results for the selected matrices in group 1, (2) the geometric mean for group 1, (3) the results for the selected matrices in group 2, (4) the geometric mean for group 2, (5) the geometric mean for the full set of 62 matrices.

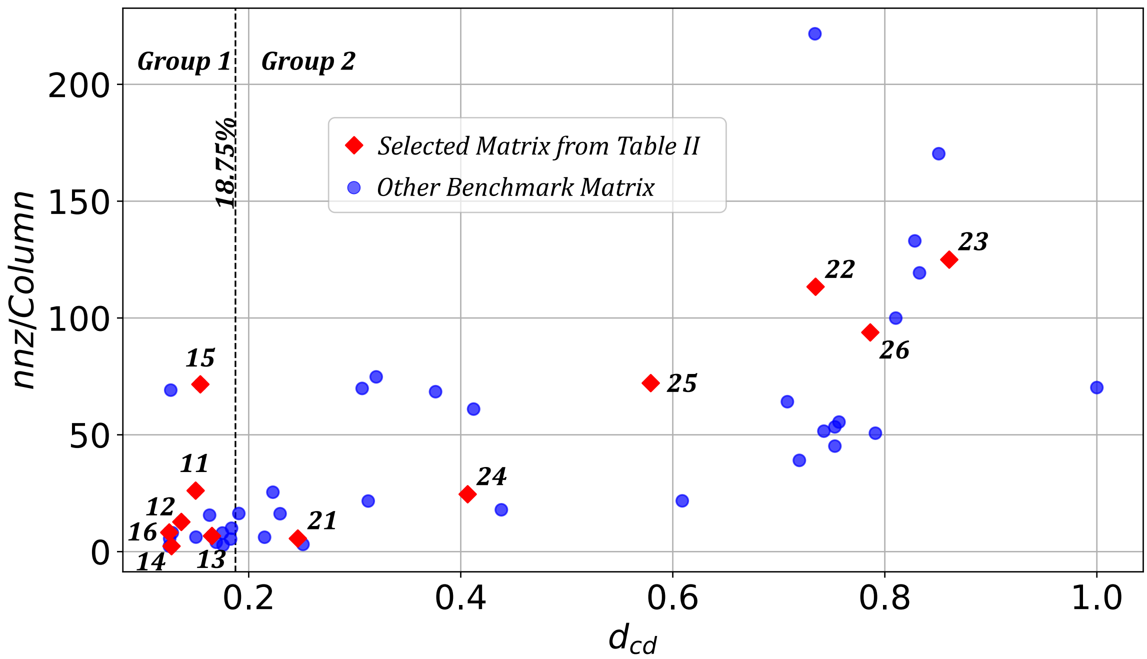


Figure 5.3: Benchmark Matrices: d_{cd} vs $nnz/Column$

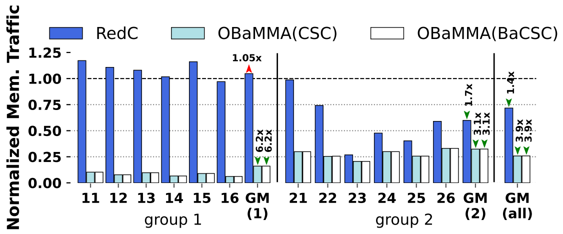


Figure 5.4: Normalized Output Memory Traffic

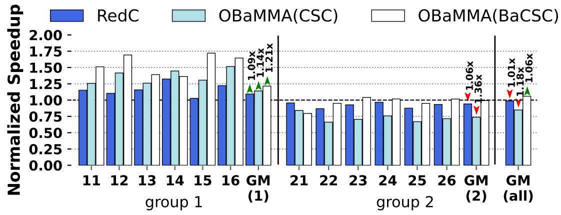


Figure 5.5: Normalized Speedup

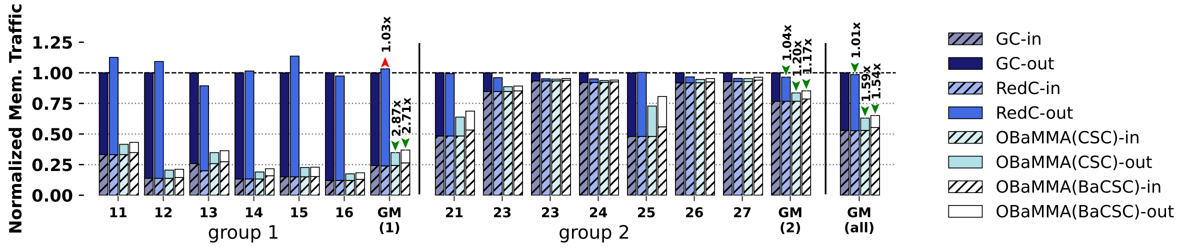


Figure 5.6: Normalized Memory Traffic on Input Matrix, Input Vector and Output Vector

5.3.3 Evaluation

For all the benchmark matrices, we simulated the SystemVerilog model executing the full **OP** SpMV and initially measured the memory traffic for the output vector as shown in Figure 5.4 for *RedC*, OBaMMA with BaCSC and OBaMMA with CSC. These results are expressed as a ratio compared to the memory traffic for *GC*. OBaMMA achieves a remarkable $3.9\times$ reduction in memory traffic which is over $2.7\times$ lower compared to *RedC*. OBaMMA performs well for the matrices in both groups. Since *GC* performs better for group 2, the relative benefit of OBaMMA is smaller but still significant. With OBaMMA, the data in the DRC is dense and stays in the cache from column to column enabling the rotating pattern of Section **??** to occur. Outside of the DRC, the opportunities for reuse are limited and the element-wise, fully-associative storage makes best use of the memory bandwidth by avoiding the transfer of mostly empty cache-lines.

Note that for the group 1 matrices, the *RedC* traffic is higher than for *GC*. Indeed, when the cache-lines are always sparsely used, the cache is ineffective, both for *GC* and *RedC*. In our configuration, the size of *GC* is $2\times$ larger than the DRC region in the *RedC*, which explains the better performance. For the group 2 matrices, the cache becomes effective, as the cache-lines frequently hit, and it becomes possible to take advantage of the in-memory accumulation provided by *RedC*.

So far, we focused on output memory traffic. To perform an **OP** SpMV, the input matrix and input vector are read once in a linear fashion [28]. Figure 5.6 shows the distribution of the total memory traffic including inputs and the output vector. We note the stark difference in behavior between the group 1 and group 2 matrices. For group 1, with traditional caches, the output traffic is high because near empty cache-lines are frequently transferred back and forth to main memory and, this is one of the reasons that **OP** algorithms are less used, despite having theoretically better performance [28, 41]. For these matrices, OBaMMA greatly reduces the output traffic providing a $2.7\times$ reduction in overall traffic. For the group 2 matrices, OBaMMA still provides a 17% reduction in overall traffic, although these matrices already perform well with traditional caches. For *GC*, *RedC* and OBaMMA with CSC, the volume of input traffic is constant. With OBaMMA, the preprocessing adds a small input traffic overhead ($< 10\%$).

Two parameters characterize sparse matrices: (1) the $nnz/Column$ which is a coarse-grain measure of density and (2) the d_{cd} which is a measure of the density at the granularity of a cache-line. In Figure 5.7, the normalized memory traffic ($bytes/nnz$) for each of the matrices is shown for *RedC* and OBaMMA. With *RedC*, the memory traffic is high for low values of d_{cd} whereas for OBaMMA, the memory traffic is virtually independent of both density parameters.

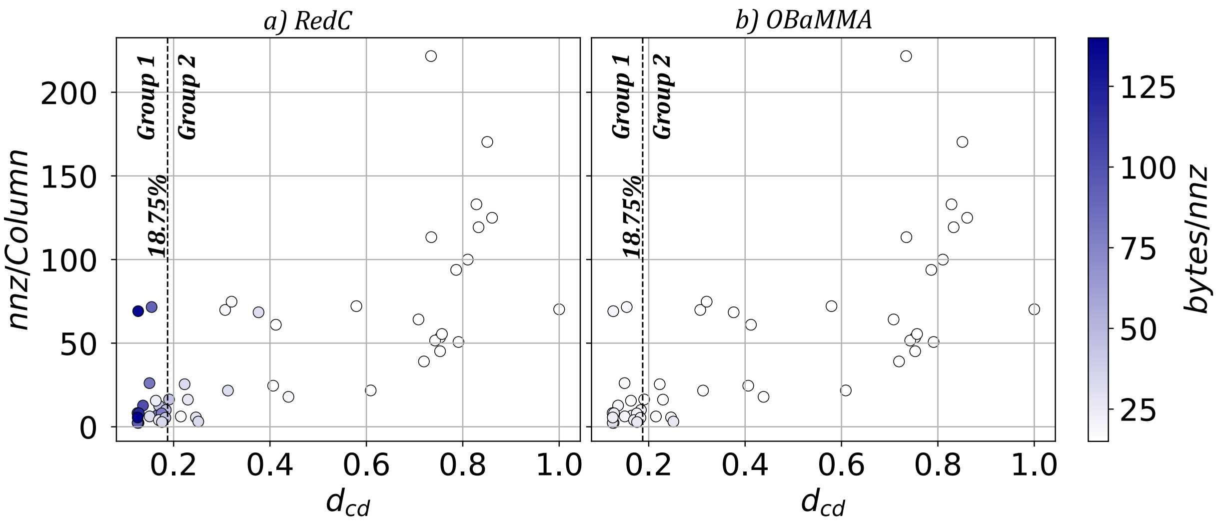


Figure 5.7: Scatter Plot of Normalized Memory Traffic for Benchmark Matrices (d_{cd} vs $nnz/Column$) in $bytes/nnz$

Our focus now shifts to the total compute time, and we recall that all main memory reads incur a fixed 64 cycle latency. Figure 5.5 shows the speedup compared to execution on *GC*. For the group 1 matrices, we first note that *RedC* achieves some speedup ($1.09\times$) which can be explained as the RED operation is "fire-and-forget" for the processor, however, they can quickly block in the cache if the *accumulation table* (*AccTab*) is busy processing the previous request. In the group 1 matrices, because the cache-lines are sparse, there are frequently bursts of evictions making this effect dominant, thus the system is memory bound. OBaMMA slightly outperforms *RedC* because it can also issue "fire-and-forget" element-wise memory transactions which reduces the blocking.

For the group 2 matrices, particularly, with our single core RISC-V processor, the system becomes compute bound. For OBaMMA with CSC, there is an overhead to calculate the region which results in some slow down ($-1.36\times$), however, with preprocessing, this overhead is removed and OBaMMA achieves the same latencies as *GC*.

To summarize, the group 1 matrices are the ones that perform poorly with traditional caches, and for this group, OBaMMA achieves a $2.4\times$ reduction in memory traffic and a speedup of $1.21\times$. In a processor with more compute resources (e.g. multicore) the pressure on the memory system would be greater and the majority of matrices would behave as the group 1 matrices. Indeed, with OBaMMA, the memory traffic is insensitive to the characteristics of the matrices as shown in Figure 5.7 which makes for predictable performance.

We previously discussed how the single-entry *AccTab* was a performance bottleneck for *RedC*. We investigate the impact of increasing the *AccTab* entries from 1 to 8, presenting the speedup relative to *GC* for selected group 1 and 2 matrices.

For the matrices in group 2, which are compute bound, increasing the *AccTab* entries has no impact on runtime ($\leqslant 1\%$). In the case of group 1 matrices, setting the *AccTab* to 1, 2, 4, and 8 entries allows *RedC* to achieve performance improvements of $1.16\times$, $1.36\times$, $1.60\times$, and $1.61\times$ respectively, compared to *GC*, although this comes at the cost of increased area. For OBaMMA, increasing the *AccTab* has virtually no impact regardless of the matrices ($\leqslant 0.5\%$). This is expected as there are rarely evictions from the DRC as it is sized to contain the band. OBaMMA's region based architecture thus reduces both total memory traffic and memory bursts.

Both HPDcache and OBaMMA systems were synthesized in GF22FDX technology using Synopsis DC™. The original area was $0.99\ mm^2$ and the OBaMMA version was $1.3\ mm^2$, an increase of 26%. This shows that OBaMMA could reasonably be integrated into a processor, providing a large performance improvement ($3.9\times$ in memory traffic), with an area overhead that is much lower than predicted by Borkar's law [**?**].

5.4 Conclusion

TODO

TAKEAWAYS

TODO

Chapter 6

Evaluation

Chapter 7

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Contents

1 TODO

TODO

Appendices

Contents

A Sparse Formats

TODO

COO The simplest *sparse format* is called *COO* (*COOrdinate*) [13, 46, 52]. It consists in three separate tables, one filled only with the non-zero elements of the matrix, a row index table containing the row of the corresponding elements, and similarly a column index table (see Figure **??**). The two latter are *metadata* tables and all three tables are of size *nnz*. In this format, the elements are not necessarily sorted, the memory footprint does not depend on the structure of the matrix and all elements are stored independently of each other.

CSR & CSC The most commonly used *sparse format* is called *CSR* (*Compressed Sparse Row*) [6, 13, 46, 52]. It is similar to the *COO* format except that the row index table is replaced by a set of indices that point to the position of the start of each row in the column index table (see Figure **??**). The size of this table is the number of rows plus one, and, in practical cases, the memory footprint is smaller than with *COO*. However, the non-zero elements need to be sorted in *row-wise* order. This format has a *column-wise* variant called *CSC* (*Compressed Sparse Column*) where the column index table stores the positions of columns instead of rows. These two formats are symmetrical but with *CSR*, it is the traversal of rows that is efficient because they are sequential whereas in *CSC*, column traversal is efficient.

BCSR Even in sparse matrices, there can be regions, or blocks, that have a large fraction of non-zero values. In this case, the *BCSR* format [6], which is an extension of the *CSR* format with *tiling*, can be attractive. We use the term *tiling* to describe a level of hierarchy in a storage format. That is, a *tile* refers to a sub-matrix, which itself may be dense, as in the case of *BCSR*, or which could be sparse. With the *BCSR* format, we simply replace the non-zero elements of the *CSR* format with dense, fixed-size sub-matrices (see Figure **??**). Within these sub-matrices, it is possible to have some zero elements. The advantage of the *BCSR* format is that the dense sub-matrices can be processed using techniques optimized for dense matrices, particularly using vector units or GPUs [27]. Compared to the *CSR* format, *BCSR* makes it possible to accelerate the computation within the dense *tiles*, but this comes with the overhead of storing some zero values. Not all matrices are well suited to the *BCSR* format.

DIA For diagonally structured matrices ($\forall(i,j) \in [\![0,M]\!] \times [\![0,N]\!],\ a_{i,j} = 0\ for\ |i-j| > K$), a format called *DIA* (*DIAgonal*) is particularly well suited [6, 13, 46]. It consists in storing the values along the $2 \times K + 1$ diagonals in a table and keeping information about the position of the diagonals in the matrix (see Figure **??**). The organization of the non-zero data and selection of the diagonals can be tailored to the matrices and access patterns. This format is well suited to diagonal matrices, has a low overhead for storing indices and can be combined with other formats to better fit more complex structures.

ELL Another format called *ELL* (*ELLPACK*) [6, 13, 32, 46] is well suited when the number of non-zero elements per row can be bounded by K. In this case, the non-zero values in a $M \times N$ matrix can be stored as a dense $M \times K$ matrix, augmented with an index table, of the same dimensions, which gives the column position of each non-zero value (see Figure **??**). This format is not efficient if the number of non-zero elements per row is highly variable, as many entries in the $M \times K$ table are unused. This format is well suited to GPUs [11, 54] since it transforms a sparse matrix into a dense one in memory.

HYB The *HYB* format (HYBrid) [7] is an example of a format which combines *ELL* and *COO*. The *ELL* format can be used, but with a value of K (the width of the value table) that is smaller than the maximum non-zero entries per row in the original matrix (see Figure **??**). For the small number of rows where the table lacks a sufficient number of columns, the *COO* format can be used to store the extra values. In this way, the storage efficiency of *ELL* is improved, and it can be applied to a broader class of matrices.

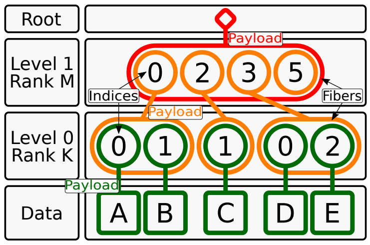

Figure 7.1: Example of the *FiberTree* Representation with CSR Format

B A Generic Representation for *Sparse Formats*

A visual representation of compressed matrices, called *FiberTree* [50], is helpful to understand these sparse formats. Each *rank* of a matrix represents a level of a tree where the nodes are filled with either the metadata of the child nodes or the matrix data which can only appear in the leaves (see Figure 7.1). With this representation, we clearly identify the number of *ranks*, which define the *fibers*. A *fiber* is the generic term to describe a one-dimensional stream of data or metadata, which corresponds to a row or column in the case of a two-dimensional matrix. A stream of indices is a *fiber* of metadata.Figure 7.1 presents, as an example, a *FiberTree* representation of the matrix of Figure **??**, matching the CSR format: each *rank* in the tree corresponds to a *fiber* stored in the table *Row PTR* or *Col IDX* of Figure 1.3b.

In a recent work [57], a systematic nomenclature for *sparse matrix (and tensor) formats* was proposed, based on *FiberTree* representation. In fact, in the example of Figure 7.1, the *fibers* are compressed with different methods. Thus each *rank* can be characterized by a label indicating how it is compressed: *Uncompressed* (*U*), *Coordinate Payload* (*CP*), *Uncompressed Offset Pairs* (*UOP*). *U* defines *ranks* where all data values are stored contiguously in memory. *CP* defines *ranks* that store the indices of each payload, a payload being either a non-zero element or a sub-matrix[1]. *UOP* defines *ranks* that store pointers (offsets) to delimit sub-matrices of the next *rank*[2]. These are the three main ones as they are used in the most common sparse formats but many others exist or can be created.

With this nomenclature, the sparse formats can be classified according to their *rank* compression methods. For example, the CSR format is denoted as *(UOP, CP) row-major* since its *Row PTR* table stores pointers (offsets) to rows. The CSR and CSC formats are both of type *(UOP, CP)*, one being *row-major*, the other *column-major*. The COO format is denoted as CP^2 because there are two coordinates (row and column) for each non-zero element, as it is a *CP fiber* having two indices per data element instead of one. User defined formats can also be easily described and named. For example, a user could define a storage format with a nested CSR. That is, each non-zero element in the outer CSR, corresponds to a sub-matrix, itself stored in CSR format. Such a representation would be labeled *(UOP, CP, UOP, CP)*. A similar nomenclature is proposed by TACO [13]. It is more adapted to code generation applications, where the *FiberTree*-based nomenclature better describes the memory storage of the sparse format. For example, *U* corresponds to *Dense*, *CP* to *Singleton*, the pair *(UOP, CP)* is translated as *Compressed* while *UOP* alone corresponds to *Offset*. We chose to adopt the *FiberTree*-based

[1] In this section, we use the term sub-matrix but this is extended to the notion of *tiling* in Section 1.4.1.
[2] If some sub-matrices are empty, pointers will be repeated, hence 'uncompressed'.

nomenclature in this article since it is more suited to a hardware context.

C Extended Overview of the State-of-the-Art Accelerators for Sparse Computing

TODO

D Sparse Matrices Used for the Evaluations

TODO: table of all 62+ matrices, with characteristics, which eval they were used for

E Analytical Model of a *Reduction Cache* Computing an `OP` SpMV Kernel

TODO

F Python Model of a Data-Cache with Support for *Reductions*

TODO

G Statistics

TODO (get infos before end of contract)
(commits, nb. of lines, bugs, timeline -> into figures?)

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Contents

Plan (Brouillon)

- Introduction **(~3 p.)**
  - "light" > give the entry point on sparse matrices and irregular accesses **(1-2 p.)**
  - Mains contributions **(1 p.)**
  - introduce the structure of the manuscript
- Background **(~5 p.)**
  - irregular accesses -> def, issues (in memory but not only?), domains/applications (including sparse matrix processing) **(1 p.)**
  - Focus on sparse matrices, context: it is widely used, in many domains, sparse, large, structures
  - Sparse matrices in a kernel: show where the accesses are regular vs irregular (infos on basic sparse formats (other sparse formats in appendix x), tiling, basic kernels, basic algos...) **(2-3 p.)**
  - Hardware challenges clearly defined (gather, scatter...) **(1 p.)**
- SoA **(~11 p.)**
  - Present solutions in SW for CPU/GPU, such as special sparse formats, possibility to play on algo (gust analysis), tiling methods adapted for sparse data (specific sparse format + specific algos) (expliquer clairement que ces solutions sont interressantes et pourquoi on se concentre sur le HW) **(1-2 p.)**
  - Present solutions in HW (dedicated accelerators) -> comparison table along common characteristics **(1-2 p.)**
  - Presentation of the major accelerators (selection of 3 important: processor assist PHI, standalone GAMMA, multi algo SPADA, ACES) (others in appendix xi) **(3-4 p.)**
  - Problem of comparison / comparison of standalone / other analysis (issue of matrix structure) -> Objectives: convince reader that SoA was done in depth, and acknoledge the matrice structure issue **(1 p.)**
  - Takeaways -> paths for designing an accelerator (end od SoA paper) **(1-2 p.)**
  - We choose a path -> show which one (coarse grain, at the level of the SoA, comparable to SortCache, PHI...)
- High level architectural presentation of SpDCache **(~10 p.)**
  - Reminder of the path we choose with more details: cache / reductions / OP SpMV / ... **(1 p.)**
  - Emphasis the comparison between concerned SoA accelerators -> show what is already done and useful, show what is missing
  - Present the case of banded matrices as an example of structure to leverage **(1-2 p.)**

  - SpDCache, architecture presentation with two regions, details on REDs transactions, IBRC vs OBRT **(7 p.)**
  - Initial results from Python model (ASAP) and note that a probabilistic model and RTL model will follow
- Mathematical analysis of the cache memory accesses for processing sparse matrices **(~23 p.)**
  - Explain the objectives of this chapter (~formal justification of the concept + cache size/region dimensioning) **(1 p.)**
  - Describe the design space of the path we choose (with reminders if necessary) -> introduce base parameters and/or variables, give the exact algorithm... **(1 p.)**
  - Show the interest of separate cache regions for A/b and c (formal) -> show that a GC is obviously not efficient when random accesses / confirm why SortCache, PHI, ... exist **(2 p.)**
  - Show that A and b are not a limiting factor with a formal reasoning -> give the needed space for GRC when prefetcher (clear statement that gather is not the aim of the thesis) **(2-3 p.)**
  - We use a GRC (no modification) for A and b -> get nb loads
  - Show the interest of a reduction cache region for c -> REDs / reduce the number of mem accesses / link to SoA (see ASAP) **(2 p.)**
  - We created a probabilistic model of the reduction cache which is based on the following principles ... (presented in detail in appendix n) / avantage of this model / we implemented this model in C / validation of the model with state of the art article on generic cache formalisation / introduce the graphs for the next sections **(1-2 p.)**
  - Show that reduction cache does not handle randomness -> model graph **(1-2 p.)**
  - Show RedC 1reg vs RedC 2reg on banded example **(2-3 p.)**
  - Show best cache dimensions for the dense part (band) **(1 p.)**
  - Show that sparse region (out of band) could use an other region for fine grain density (case of sub bands) -> dimensions **(1-2 p.)**
  - Show that sparse region could use adapted HW because of the isolated elements -> dimensions **(1-2 p.)**
  - Show how it would work with a uniform matrix -> how does it affects the dimensions **(1 p.)**
  - Conclude on our solution and dimensioning **(1 p.)**
- Micro architectiture of SpDCache **(~10 p.)**
  - Present our hardware solution, based on the concept
  - Start with basic info on HPDcache (only the one that are useful for SpDCache)
  - Describe modif in pipeline / hazard management / throughput
  - Modif in RAM accesses ...
  - Stats: git / bugs / lines of code / synthesis
- RTL simulation results **(~10 p.)**
  - Experimentations: RTL simulation results -> matrices chosen, generated / metrics / ...
  - Results / comparison with theory / analysis / conclusion
- Future enhancement **(~8 p.)**
  - Architecture presentation with three regions (add OBRC, CB...) **(2 p.)**
  - Architecture of eviction anticipation **(2 p.)**

  - Present multicore version -> scale **(2 p.)**
  - Present additional optimisation (region config, learn from first iteration) **(1-2 p.)**
- Conclusions **(~2 p.)**
  - ..
- Appendices
  - All sparse formats **(~5 p.)**
  - Other accelerators of SoA **(~10 p.)**
  - Probabilistic model of the reduction cache **(~10 p.)**
  - Python model **(~1 p.)**